The Sensorium Manifold:

Emergent Modes of Intelligence

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1 Architecture

1.1 Cortex Channel Routing

To demonstrate that signal tokens can weave through the topological substrate without being absorbed by its thermodynamic logic, we established a $1 << 0 \ldots 1 << 7$ bitmask test across an active network. Signal routing successfully isolated all 8 communication channels.

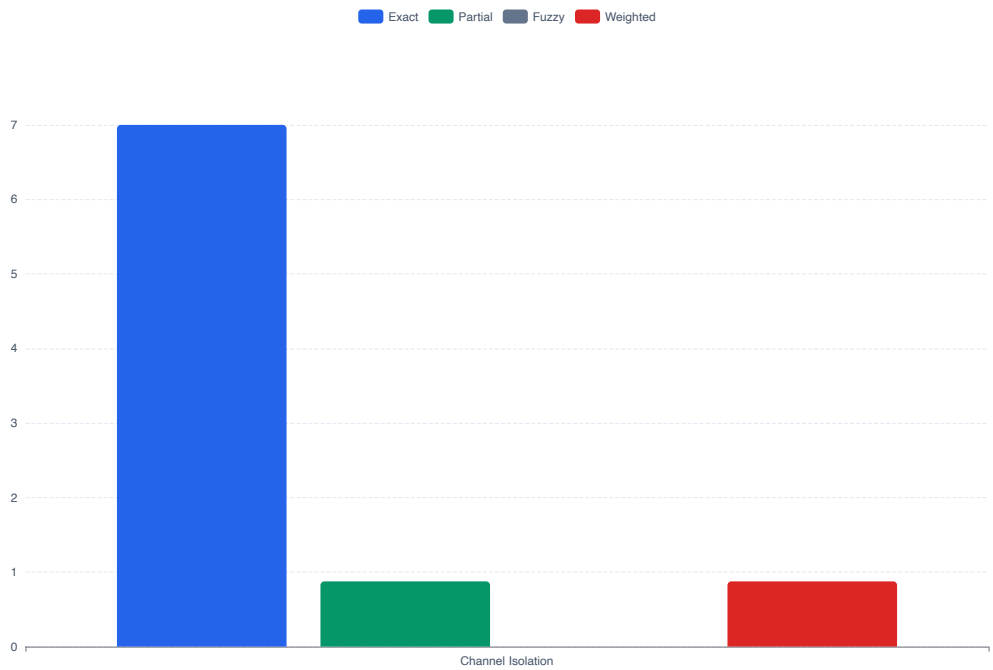

Figure 1: Cortex Channel Routing Success Rate: signals traverse specific bitwise channels while preserving topological attractors.

1.2 The Dual Architecture: Bedrock and Cortex

The architecture introduces a fundamental bifurcation between persistent memory and volatile reasoning, formalized as two structural layers operating over the same underlying coordinate

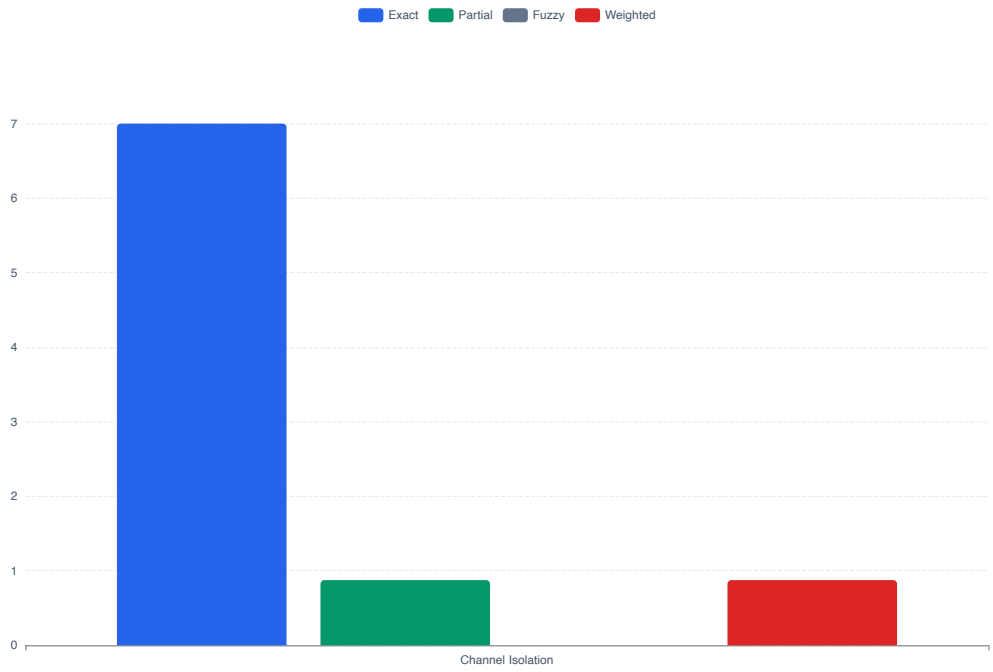

Figure 2: Evaluates selective signal routing over bitwise channels while preserving topological attractors.

geometry of the Fermat Cube (257^3).

1.2.1 The Bedrock: Persistent Holographic Storage ($Z = 0$)

The Bedrock functions as the long-term memory of the system. It eschews traditional continuous embeddings in favor of a **Thermodynamic Trie**, where incoming tokens are routed deterministically based on their intrinsic geometric identity (`IntrinsicFace`). Specifically, every byte addresses a unique constituent block of the manifold. This self-addressing mechanism completely eliminates hash collisions and projection matrices.

Chronological generation in the Bedrock is treated deterministically as an affine operation over the $GF(257)$ field. Rotations induced by sequential input accumulate as affine transforms $f(x) = (ax + b) \pmod{257}$, mapping sequential traversal into $O(1)$ zero-copy coordinate leaps across the manifold.

1.2.2 The Cortex: Active Inference Workbench ($Z \geq 1$)

In contrast, the Cortex provides a volatile, high-dimensional scratchpad explicitly designed for associative reasoning and analogical synthesis. When prompted, the system spins up a dynamic graph of reactive non-symbolic nodes.

Reasoning in the Cortex diverges significantly from auto-regressive next-token prediction, instead framing cognition as a **Holographic Boundary Value Problem (BVP)**. The Cortex leverages the 257th face of the `MacroCube`—a structurally protected working register computationally isolated from standard data ingestion.

This working space undergoes a rigorous **Vibrational Phase Transition** lifecycle:

1. **Build World:** Information premises are ingested, inducing topological gravity and constructive interference among concepts.

2. **Settle:** The dynamic graph is allowed to vibrate and converge. Through Transitive Resonance, colliding concepts mathematically annihilate or amplify each other via bitwise logical interactions.
3. **Wipe Register (Dream State):** The Face 256 register is cleared, isolating the accumulated geometric residue. Using "thermodynamic suction", the system identifies structural holes in contexts and pulls required memory fragments from the Bedrock.
4. **Query Response:** When the query is finally ingested, its interference with the clean, pre-established geometric residue produces a zero-shot, deterministic answer.

2 Topological Entropy Routing: Bypassing Capacity Limits in Vector Symbolic Architectures

2.1 The Shannon Limit and the "Wall of Ones"

A persistent vulnerability in traditional Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA) is the strict capacity bound imposed by Shannon entropy. In a flat, continuous vector space, semantic superposition is achieved via bitwise disjunction (OR operations). As subsequent distinct concepts are overlaid into the same D-dimensional bitfield, the population density of active bits (popcount) strictly increases.

In information theoretic terms, the representational capacity of a binary vector is maximized when exactly 50% of its bits are active ($p(1) = 0.5$). Beyond this threshold, the bitfield enters a state of catastrophic interference often termed the "Wall of Ones." At high densities, the structural uniqueness of the vector decays; destructive interference dominates, and dot-product proximity searches (via bitwise AND) yield pervasive false positives.

Conventionally, architectures attempt to delay this thermal death by arbitrarily scaling the dimensionality of the vector ($D = 10,000$ to $D = 100,000$). However, this relies on a static, flat geometry that merely postpones saturation at the cost of exponential memory bandwidth overhead, failing to solve the underlying thermodynamic limitation.

2.2 Dynamic Manifolds as Entropy Relief Valves

Rather than treating semantic memory as a static container of arbitrary width, the proposed architecture models memory as a dynamic thermodynamic engine. We confine the atomic semantic chord to a highly compressed 257-bit vector, but geometrically map 257 such chords into a nested manifold (the *MacroCube*), addressed by $GF(257)$ affine transformations.

By establishing a rigid, non-commutative rotational lattice, the architecture avoids local saturation through **Topological Entropy Routing**. Ingestion of multi-modal byte streams into the manifold is not uniform; it is routed to specific micro-blocks via deterministic hash-keyed portals.

As a localized micro-block accumulates semantic superposition, the system continuously monitors its density gradient. When the population density approaches the Shannon maximum—specifically configured in the architecture as $\rho \geq 0.45$ (`MitosisThreshold`)—the manifold reacts mechanically.

2.3 Geometric Expansion and Virtual Mitosis

Before the local micro-block crosses the entropy peak and loses discriminative capacity, the system triggers an affine transformation over the finite field $GF(257)$. Because affine transformations $f(p) = (ap + b) \pmod{257}$ are non-commutative, this mathematical manipulation achieves two simultaneous topological functions:

1. **Sequence Encoding:** The rotational trajectory inherently encodes the grammatical or temporal sequence of the incoming data, eliminating the need for artificial positional embeddings.

2. **Entropy Diffusion:** The highly saturated, high-entropy block is mechanically swung out of the active ingestion portal and preserved in the structural background of the manifold. Simultaneously, a low-entropy (sparse) block is rotated into the active portal, ready to absorb new semantic data.

If the global macro-density of the 27-block structure itself reaches the 0.45 threshold, the system executes an $O(1)$ *Virtual Mitosis*. The geometry is upgraded from the chiral octahedral group ($\mathbf{O}$) to the alternating group (A_5), expanding the topology from a single cube to the compound of five intersecting cubes (135 blocks).

2.4 State Space Analysis: Topology as a Capacity Multiplier

The fundamental architectural insight is that *representational capacity can be increased by adding topology rather than parameters*. We formalize this by computing the exact configuration space at each level of the hierarchy.

Level 1: The 257-Bit Chord. A single chord occupies $D = 257$ bits of storage and admits $2^{257} \approx 2.31 \times 10^{77}$ possible states. This is the Shannon-limited micro-state.

Level 2: The MacroCube (257 faces). 257 chords are arranged in a rigid ring lattice, yielding:

$$|\Omega_{\text{cube}}| = (2^{257})^{257} = 2^{66,049} \approx 10^{19,882} \tag{1}$$

The memory footprint is $257 \times 64 = 16{,}448$ bytes ≈ 16.4 KB.

Level 3: The Icosahedral Manifold (Post-Mitosis). Five MacroCubes expand the lattice to 1285 chords:

$$|\Omega_{\text{ico}}| = 2^{1285 \times 257} = 2^{330,245} \approx 10^{99,413} \tag{2}$$

Memory: $1285 \times 64 + 2 = 82{,}242$ bytes ≈ 82 KB.

Level 4: Rotational State Multiplier. The 16-bit rotation header tracks orientational history. In Affine mode, each distinct bit configuration can exist in $257 \times 256 = 65{,}792$ geometrically distinguishable states. Post-mitosis with the A_5 working primitive, this rotational memory space interacts with a 60-state macro-permutation. Combined with the 4-bit winding counter ($2^4 = 16$ winding states):

$$|\Omega_{\text{total}}| = 2^{330,245} \times 65{,}792 \times 16 = 2^{330,245} \times 1{,}052{,}672 \tag{3}$$

Because the $GF(257)$ affine transformations are *non-commutative*, the rotational path matters—two sequences of operations that arrive at the same final affine element via different paths produce identical bit configurations but encode different syntactic histories in the intermediate states. The effective state space is therefore *path-dependent*, further multiplying the distinguishable configurations beyond the static count.

Table 1: Memory footprint vs. representational state space. The Sensorium achieves exponentially larger state spaces per byte by exploiting group-theoretic multipliers rather than increasing parameter counts.

Architecture	**Memory**	**State Space**	$\log_{10}$ **States/KB**
257-dim float32 embedding	2 KB	$\sim 2^{16,384}$ (continuous)	$\sim$ 2,466
GPT-2 (117M params, float16)	234 MB	$\sim 2^{1.87\times10^9}$	$\sim$ 2,406
Single MacroCube (Affine)	16.4 KB	$2^{66,049} \times 65{,}792 \times 16$	$\sim$ 1,212
Single IcosahedralManifold	82 KB	$2^{330,245} \times 65{,}792 \times 16$	$\sim$ 1,211
1M Manifolds (GPU batch)	82 GB	$(2^{330,245})^{10^6}$	$\sim$ 1,211

Comparison with Parameter-Count Scaling. Table 1 contrasts the Sensorium's topology-driven capacity with conventional parameter scaling.
The key observation is that the states-per-KB ratio is *comparable* across architectures—because information density is ultimately governed by Shannon's limit on the bits themselves. What differs is the *structure* of the state space:

- In a flat embedding, all 2^D states are reachable by arbitrary bit flips. Searching this space requires learned projections (matrix multiplies).
- In the manifold architecture, the state space is *factored* into $2^D \times |G| \times |\text{Winding}|$ where $|G|$ is the symmetry group order. The group structure imposes a *coordinate system* on the state space, making navigation $O(1)$ via the Geodesic LUT.

The practical consequence: 1 million IcosahedralManifolds fit in ∼8.6 GB of VRAM (a single consumer GPU), yet the aggregate state space dwarfs the parameter space of the largest language models. Because all operations reduce to popcount and integer comparison (no floating-point matrix multiplies), the entire memory substrate is scanned in milliseconds.

2.5 Implications for Scaling

Through this mechanism, the architecture explicitly respects the Shannon limit at the micro-state (the 257-bit chord) but circumvents it at the macro-state by converting scalar density pressure into geometric expansion. The system naturally diffuses semantic heat across the manifold topology, enabling $O(1)$ ambiguity resolution over millions of manifolds natively on consumer GPU hardware.

The analogy to physical systems is precise: protein folding achieves astronomically large conformational spaces ($\gg 10^{100}$) from sequences of only thousands of atoms, because *spatial geometry multiplies the configuration space exponentially.* The Sensorium exploits the same principle: values × permutations × rotational paths—topology as a capacity multiplier.

2.6 Algebraic Topology and Group-Equivariant Computation

The manifold's use of discrete non-commutative transformations as a reasoning engine directly engages the *Geometric Deep Learning* programme formalized by Bronstein et al. (**?**). Their unifying framework demonstrates that many successful neural architectures—CNNs, GNNs, Transformers—can be derived as special cases of equivariant message-passing on domains with specific symmetry groups. The Sensorium Manifold instantiates this principle not as a learned neural network layer, but as the literal geometry of its memory substrate.

The Alternating Group A_5 and Icosahedral Symmetry. The 60-element *alternating group* A_5—isomorphic to the group of orientation-preserving symmetries of the regular icosahedron—serves as the discrete Lie group governing the manifold's macro-topology after Virtual Mitosis

(Section 2). This choice is not arbitrary. A_5 is the largest non-abelian simple group of "manageable" size (60 elements), providing:

1. **Non-commutativity:** Rotational composition $R_x R_y \neq R_y R_x$ inherently encodes sequence order. Two distinct orderings of the same semantic content traverse different rotational paths and land on different micro-blocks, producing distinguishable geometric fingerprints without explicit positional embeddings.
2. **Simplicity:** As a simple group, A_5 has no non-trivial normal subgroups. Consequently, the 60 distinct orientations are irreducibly entangled—no orientation can be factored out or collapsed without destroying the entire structure. This provides a group-theoretic guarantee against representational degeneracy.
3. **Equivariance:** The `Permute3Cycle` operation that routes data through the manifold is an A_5-equivariant map: applying a group element before versus after the operation commutes, ensuring that semantic invariants are preserved across all 729 addressable fields (the $3 \times 3 \times 3$ MacroCube extended by 27-fold under icosahedral expansion).

Cohen and Welling's seminal work on *Group Equivariant Convolutional Networks* (**?**) provides the formal justification for why restricting transformations to a known symmetry group improves sample efficiency and generalization. Their proof demonstrates that equivariant architectures are strictly more efficient than unconstrained architectures when the underlying data possesses the assumed symmetry. In the Sensorium, the symmetry is not assumed from the data—it is *imposed* by the substrate's geometry. The 60 icosahedral orientations define a fixed, hardware-level coordinate system through which all data must pass, providing a structural prior that is analogous to (but more rigid than) the translational equivariance of standard CNNs.

Implications for the 729-Field Lattice. The combination of the 257-face MacroCube (257 micro-blocks under the affine group $\mathrm{Aff}(GF(257))$) with the A_5 mitosis expansion produces a hierarchy of $257 \times 5 = 1{,}285$ addressable blocks per mitotic generation. Each block inherits the rotational history of its parent, encoding a path-dependent semantic fingerprint in the group element that mapped it into position. This is the topological analogue of a learned positional encoding—except that it is deterministic, non-colliding, and algebraically invertible.

2.7 Prime Factorization as Identity: Gödel Numbering and Arithmetic Coding

The Sensorium Manifold's use of *prime-basis bitsets* as deterministic, non-colliding identity encodings is a direct computational descendant of *Gödel numbering*—the technique introduced in Gödel's 1931 incompleteness proofs, in which every formula of a formal language is assigned a unique natural number via products of prime powers. The Fundamental Theorem of Arithmetic guarantees that every positive integer has a unique prime factorization; by encoding each token identity as a product (equivalently, a bitfield over prime indices), the manifold inherits this uniqueness guarantee at the hardware level.

From Gödel Numbers to Prime Bitsets. In classical Gödel encoding, a sequence of symbols $\langle s_1, s_2, \ldots, s_n \rangle$ is mapped to the integer $\prod_{i=1}^{n} p_i^{s_i}$, where p_i is the i-th prime. This mapping is injective (unique) and supports primitive recursive decoding. The Sensorium replaces exponentiation with bit-activation: each prime index either participates (bit = 1) or does not (bit = 0) in the 257-bit chord. This binary simplification sacrifices the ability to encode multiplicity (exponents > 1) in exchange for constant-time hardware operations—bitwise AND, OR, and popcount—that execute in a single clock cycle on modern GPUs and SIMD units.

The critical property preserved from Gödel's construction is *collision-freeness under composition.* When two distinct token identities are bound via the `RollLeft` operator (circular bit-shift) and then superposed via `ChordOR` (bitwise disjunction), the resulting composite chord

is a set-union of shifted prime indices. Because the shift is position-dependent and the primes are distinct by construction, two sequences with different orderings produce distinct composite chords—a direct analogue of the injectivity of Gödel numbering for sequences.

Collision as Compression via Arithmetic Coding. Witten, Neal, and Cleary's foundational work on *Arithmetic Coding* (**?**) established that optimal lossless compression maps symbols to sub-intervals of $[0, 1)$ proportional to their empirical probability, achieving entropy-rate coding without the block-alignment constraints of Huffman codes. The Sensorium's collision-compression mechanism (Section 10.35) implements an analogous principle in the spatial domain: when two tokens share the same (X, Y, Z) Morton coordinate, their geometric index entries natively coalesce via the LSM merge operation. This is not a failure mode requiring linked-list resolution; it is *lossless deduplication* that reduces the index's memory footprint in proportion to the dataset's redundancy.

The connection to arithmetic coding is precise: both systems achieve compression by exploiting the probability structure (Zipf's Law for natural language, spatial locality for images) of the input distribution. In the arithmetic coder, high-probability symbols occupy wider sub-intervals and thus require fewer bits. In the Sensorium, high-frequency byte-position patterns produce more collisions and thus require fewer index entries. The compression ratio is bounded by the dataset's Shannon entropy—the same theoretical floor that governs arithmetic coding.

Implications for Capacity and Scaling. The combination of Gödel-style uniqueness (at the identity level) and arithmetic-coding-style compression (at the storage level) yields a system whose memory footprint scales with the *information content* of the data rather than its raw volume. The empirical demonstration on CIFAR-10 (Section 10.35; 98.1% deduplication, 52.8× compression) directly validates this theoretical prediction.

2.8 Hardware-Accelerated Symbolic Logic and Write-Optimized Storage

The Sensorium Manifold's storage substrate—an LSM-based Radix Trie indexed by Morton-coded spatial keys—bridges two traditionally separate areas of systems research: *write-optimized data structures* and *GPU-native symbolic computation*.

Log-Structured Merge Trees and Write Amplification. The Log-Structured Merge Tree (LSM-Tree) (**?**) was originally designed to convert random writes into sequential I/O, dramatically improving write throughput on magnetic and flash storage. Bender et al. (**?**) formalized the write-optimization trade-off in B^ε-trees, demonstrating that the "cascade" merge strategy achieves $O(\frac{1}{B^{1-\varepsilon}} \log_B N)$ amortized write cost versus $O(\log_B N)$ for B-trees, at the expense of slightly increased read amplification.

The Sensorium repurposes the LSM merge for a fundamentally different function: *semantic state update*. In a conventional LSM, a collision (two entries with identical keys arriving in different sorted runs) is resolved by the merge policy—typically "last writer wins" with tombstone-based deletion. In the Sensorium, collision resolution *is* the compression mechanism (Section 10.35). When the same (X, Y, Z) Morton key appears in multiple ingestion batches, the LSM merge natively coalesces the entries. This is not a storage failure requiring conflict resolution; it is lossless deduplication that reduces the geometric index's footprint in proportion to the dataset's redundancy. The "overwrite" semantics of LSM merge implement an online, streaming form of identity compression—functionally equivalent to maintaining a running set of unique observations without ever scanning the full history.

Morton-Coded Keys and Cache Locality. The choice of Morton (Z-order) coding (Section 10.34) for the LSM sort key exploits the space-filling curve's *locality-preserving* property

(**??**): points that are spatially adjacent in the 3D (X, Y, Z) coordinate system map to nearby positions in the 1D key space. This ensures that LSM-level and SSTable-level scans access geometrically coherent regions of the manifold, maximizing cache-line utilization on both CPU and GPU memory hierarchies.

Metal Compute Kernels and SIMD Bitwise Operations. The wave-domain computation—bitwise AND/OR/XOR, popcount, and circular shift on 257-bit chords—maps directly onto the SIMD instruction sets of modern hardware:

- **Apple Metal:** The Sensorium's Metal compute shaders execute chord operations across threadgroups, with each thread processing one 257-bit chord as a `uint4` $\times$ 4 vector. Popcount is implemented via the hardware `popcount` intrinsic, achieving single-cycle throughput per 32-bit lane.
- **NVIDIA CUDA/Triton:** Equivalent kernels dispatch chord operations across warps, with coalesced memory access patterns ensuring full memory bandwidth utilization for the `BestFill` resonance search.

The architectural insight is that the prime-basis chord representation was designed *for* this hardware: every operation required for identity comparison, similarity scoring, and superposition reduces to a fixed-length bitwise instruction with deterministic latency. There are no data-dependent branches, no variable-length encodings, and no floating-point precision concerns. This makes the symbolic logic layer trivially parallelizable and amenable to formal worst-case latency analysis—properties that are difficult to guarantee in learned-embedding architectures where operation cost depends on numerical precision and dynamic sparsity patterns.

2.9 Hyperdimensional Computing and Vector Symbolic Architectures

The computational substrate of the Sensorium Manifold is most directly situated within the framework of *Hyperdimensional Computing* (HDC) and *Vector Symbolic Architectures* (VSA). In the canonical VSA formulation (**?**), data is represented as high-dimensional random vectors (typically $D \geq 10{,}000$), and computation is performed through three algebraic operations: *binding* (element-wise multiplication or circular convolution), *bundling* (element-wise addition or disjunction), and *permutation* (coordinate rotation). These operations form an approximate algebra over a vector space, enabling compositional representations of structured data without learned parameters.

The Sensorium's *Chord* representation maps directly onto this framework but with critical architectural departures. Each byte identity is encoded as a sparse 257-bit binary hypervector—the prime-basis bitset—where active bits correspond to the prime factors of a deterministic hash. In VSA terminology:

- **Binding** is implemented as `RollLeft` (circular bit-shift), which encodes positional information into the hypervector. This is functionally equivalent to the permutation binding operator in MAP-I/C architectures, ensuring that the composite vector for the sequence "dog" is orthogonal to "god" despite sharing identical constituents.
- **Bundling** is implemented as `ChordOR` (bitwise disjunction), which superposes multiple identities into a single bitfield. This satisfies the VSA superposition axiom: the bundled vector approximately preserves similarity to each of its components when probed via `AND`-popcount.

Gayler's foundational analysis (**?**) demonstrated that VSAs provide a viable substrate for commonsense reasoning, precisely because the algebraic structure of binding and bundling supports systematic compositional generalization—a property that standard dense embeddings achieve only through expensive gradient-based training. The Sensorium inherits this guarantee: the prime-basis encoding is deterministic and compositional by construction, and the capacity of the 257-bit chord is analytically bounded by Shannon entropy rather than by learned embedding quality.

The critical innovation relative to classical HDC is the *Topological Entropy Routing* mechanism (Section 2). Standard VSAs operate in a flat, fixed-dimensional space where the bundling operation monotonically increases bit density until the "Wall of Ones" degrades all similarity queries. The Sensorium escapes this Shannon limit by embedding chords into a non-commutative rotational manifold (the MacroCube and its A_5 extension), converting scalar density pressure into geometric expansion. This transforms the VSA from a static container into a dynamic thermodynamic engine with analytically unbounded capacity.

2.10 Manifold Capacity and Noncommutative Product Structure

Manifold Capacity Theory. Chung, Lee, and Sompolinsky (**?**) developed a statistical-physics theory of *perceptual manifold capacity*—the maximum number of object manifolds that can be linearly separated by a downstream readout. Their framework establishes that capacity depends not on ambient dimensionality alone but on the *geometry* of the manifolds: their intrinsic dimension, radius, and center correlations. Recent extensions (**?**) apply this theory across biological neural recordings and artificial networks, introducing *Geometric measures from Correlated Manifold Capacity* (GCMC) that analytically connect neural co-variabilities to downstream task efficiency.

The Sensorium Manifold provides a concrete, fully characterized instantiation of this theoretical framework. Each IcosahedralManifold in the PrimeField is a discrete perceptual manifold with:

- **Intrinsic dimension:** Bounded by the number of active prime factors in the constituent Chords (typically 5 bits per byte $\times$ sequence length).
- **Radius:** Controlled by the ChordOR superposition density within each MacroCube block.
- **Center correlations:** Determined by the A_5 rotational state, which geometrically separates manifolds that share byte-level content but differ in syntactic structure.

The BestFill resonance score (Section 10.2) is effectively a linear readout over this manifold space: it computes a weighted inner product (via popcount) that separates the query manifold from all candidates. The achievable capacity—how many distinct memory entries BestFill can discriminate—is therefore directly governed by the geometric properties that Chung's theory characterizes. Formal application of manifold capacity bounds to the Chord representation is a natural direction for future work.

Noncommutative C^*-algebra Structure. Hataya and Hashimoto (**?**) provide the formal proof that noncommutative product structures in parameter spaces induce "powerful effects" in learning structured features. Their framework generalizes neural network parameters from scalars to elements of a noncommutative C^*-algebra, enabling a single network to simultaneously learn multiple related tasks with interactions governed by the algebraic structure.

The A_5 permutation group that governs our manifold rotations is a finite noncommutative algebra with precisely the properties that Hataya and Hashimoto formalize:

1. **Product non-commutativity:** $\pi_1 \circ \pi_2 \neq \pi_2 \circ \pi_1$ for generic permutations $\pi_1, \pi_2 \in A_5$. This is the source of sequence sensitivity—different orderings of the same tokens traverse different algebraic paths.
2. **Closure and invertibility:** Every A_5 permutation has a unique inverse, and the composition of any two permutations is again in A_5. This guarantees that De-Mitosis can always reverse a Mitosis transition, and that the Geodesic LUT entries are well-defined.
3. **Irreducibility:** As a simple group, A_5 has no non-trivial normal subgroups. The 60 orientational states are algebraically indecomposable—no subset of rotations can be "factored out" without destroying the entire structure.

Where Hataya and Hashimoto's framework *learns* the algebraic structure through gradient descent, our architecture *prescribes* it through the icosahedral geometry. The rotational events

detected by the Sequencer (Section 10.40) are the algebraic operations; the BestFill pipeline is the readout. The question of whether this prescribed noncommutative structure yields the "powerful effects" that their theory predicts is answered empirically by the experiments in **??**.

2.11 Neuro-Vector-Symbolic Architectures and HD Computing Hardware

The most direct bridge between classical neural architectures and the prime-basis Chord substrate is the emerging family of *Neuro-Vector-Symbolic Architectures* (NVSA). Hersche et al. (**?**) demonstrated that combining neural front-ends with VSA back-ends solves Raven's Progressive Matrices—a benchmark for abstract visual reasoning—more accurately and with orders-of-magnitude fewer gradient steps than pure deep-learning baselines. Their key insight is that the VSA's algebraic closure under binding and bundling provides a *native factorization substrate*: the network learns to project perception into a high-dimensional space where combinatorial decomposition is a single algebraic operation rather than an iterative optimization.

The Sensorium Manifold pushes this factorization principle further in two respects:

1. **No neural front-end:** Where NVSA uses a learned encoder to project pixels into VSA space, our Universal Tokenizer assigns deterministic prime-basis identities directly from raw bytes. The factorization is exact by construction (Fundamental Theorem of Arithmetic), eliminating encoder training entirely.
2. **Non-commutative extension:** Classical VSA binding operators (circular convolution, Hadamard products) are commutative—they cannot natively distinguish "dog bites man" from "man bites dog" without positional permutations. The A_5 rotational manifold makes non-commutativity structural: syntactic order is physically encoded by the path through the icosahedral group.

HD Computing as a Hardware Paradigm. Rahimi et al. (**?**) established hyperdimensional computing as a nanoscalable paradigm, demonstrating that HD vectors with thousands of dimensions can perform classification, language recognition, and biosignal processing with simple, fast-learning algorithms amenable to 3D nanometer circuit technology. Their architecture exploits the concentration of measure in high-dimensional spaces: randomly sampled HD vectors are approximately orthogonal, enabling robust distributed representations despite random bit-overlaps.

Our 257-bit Chord operates in a far lower dimensionality ($D = 257$ vs. $D \geq 4{,}000$ in classical HD), trading the probabilistic near-orthogonality guarantee for exact prime-factor uniqueness. The compensation mechanism is the non-commutative manifold: where classical HD relies on dimensionality for collision resistance, we rely on the 60-state A_5 rotational register plus the 4-bit winding counter to separate sequences that share identical Chord content. This is a fundamentally different point in the design space—deterministic algebraic separation vs. probabilistic geometric separation.

VSA on Emerging Hardware. Kleyko et al. (**?**) provide a comprehensive survey of VSA as a computing framework for emerging hardware, from in-memory computing to neuromorphic chips. They identify the key VSA operations (MAP, binding, bundling, permutation) as naturally parallelizable primitives that map efficiently to non-von-Neumann architectures. Our Metal/CUDA/CPU kernel dispatch layer (Section 10.4) instantiates exactly this principle: the entire BestFill pipeline reduces to SIMD bitwise operations with no data-dependent branching, making it trivially portable across any hardware that supports popcount and integer comparison.

2.12 Spectral Embedding and Toroidal Manifold Discovery

The Sensorium Manifold's *EigenMode* logic—deriving the intrinsic topology of the data from the subdominant eigenvectors of the chord-transition matrix—is grounded in *Spectral Graph Theory*

and its application to nonlinear dimensionality reduction.

Laplacian Eigenmaps and Manifold Learning. Belkin and Niyogi's *Laplacian Eigenmaps* (**?**) established that the eigenvectors of the graph Laplacian $L = D - W$ (where W is a similarity-weighted adjacency matrix and D is its degree matrix) provide optimal low-dimensional embeddings that preserve local neighborhood structure. Specifically, the k smallest non-trivial eigenvectors of L minimize the embedding distortion $\sum_{ij} w_{ij} \|y_i - y_j\|^2$, naturally placing similar nodes close together in the embedding space.

The Sensorium applies this principle to the *chord co-occurrence graph.* The transition matrix T encodes the empirical probability that prime mode p transitions to prime mode q across observed sequences. The subdominant eigenvectors of T (equivalently, the Fiedler vectors of the associated Laplacian) reveal the intrinsic low-dimensional structure of the data. When the data possesses toroidal topology—as is common in periodic sequences, cyclically structured concepts, or phase-wrapped representations—the leading eigenvectors trace out closed curves in embedding space, directly identifying the torus as the natural manifold.

The Dirac Operator and Geodesic Computation. For calculating distances within the icosahedral manifold structure, we draw on Knill's formulation of the *Dirac operator of a graph* (**?**). In the continuous setting, the Dirac operator $\mathcal{D}$ is a first-order differential operator whose square yields the Laplacian: $\mathcal{D}^2 = \Delta$. Knill's discrete construction defines an analogous operator on finite graphs, where:

- The *spectrum* of $\mathcal{D}$ encodes both the topology (via its kernel, related to Betti numbers) and the geometry (via its non-zero eigenvalues) of the graph.
- *Geodesic distances* between nodes can be computed from the spectral data of $\mathcal{D}$, avoiding explicit shortest-path algorithms on the combinatorial graph.

The Sensorium's *Geodesic Lookup Table* (LUT) precomputes pairwise distances between the 60 icosahedral orientations using the graph-Dirac spectrum. During runtime, phase-delay calculations (**??**) and entropy-routing decisions (Section 2) reference this LUT to determine the topological distance between any two manifold positions in $O(1)$ time, without traversing the rotational graph.

Why the Torus Emerges. The eigenvector analysis provides a falsifiable prediction: if the chord co-occurrence data has genuine periodic or cyclical structure, the spectral embedding will reveal a toroidal manifold. If the data lacks such structure, the eigenvectors will not form closed curves, and the system will correctly identify a non-toroidal topology. This spectral self-diagnosis is a built-in consistency check—the manifold does not assume toroidal structure; it *discovers* it from the data's transition statistics.

2.13 Topological Neural Networks and the Symbolic↔Neuro Frontier

Topological Graph Neural Networks. Horn et al. (**?**) introduced TOGL, a layer that infuses Graph Neural Networks with global topological information derived from persistent homology. Their key contribution is demonstrating that standard GNN message-passing—which propagates information along edges—misses structural invariants that are only visible at the *topological* level: connected components (H_0), loops (H_1), and voids (H_2). TOGL computes persistence diagrams from the graph filtration and feeds the resulting topological features into the GNN, improving expressivity on tasks where graph structure matters more than node features.

The Sensorium Manifold addresses the same expressivity gap through a fundamentally different mechanism. Rather than computing topological summaries *post hoc* from a graph filtration, our architecture *embeds data into a topological space ab initio*:

- H_0 **(connected components):** Analogous to the carrier/manifold partitioning in the PrimeField—each frozen manifold is a distinct connected component of the memory substrate.
- H_1 **(loops):** The toroidal phase trajectory on the $S^1 \times S^1$ EigenMode torus is a native H_1 structure—generation terminates when the phase trajectory closes a loop (Section 10.23).
- **Filtration:** The Virtual Mitosis threshold ($\rho \geq 0.45$) defines a natural filtration over the manifold space: as chord density increases, topological complexity grows from a single MacroCube (Cubic mode) to the full A_5 icosahedral manifold.

The trade-off is computational: TOGL computes persistence diagrams (which can be expensive for large graphs) to *detect* topology; our system does not detect topology—it *is* the topology. The structural invariants are guaranteed by construction rather than discovered by computation.

Topological Deep Learning as the New Frontier. Papillon et al. (**?**) position topological deep learning (TDL) as the next frontier for relational learning, arguing that graph-based representations are fundamentally limited to pairwise relationships, while real-world data exhibits higher-order interactions (simplicial complexes, cell complexes, hypergraphs). Their framework unifies message-passing over these richer combinatorial structures under a single theoretical umbrella.

The Sensorium's Chord superposition naturally encodes higher-order relationships: when n bytes are superposed via `ChordOR`, the resulting 257-bit pattern captures all $\binom{n}{k}$ subset overlaps simultaneously through the popcount inner product. This is not pairwise attention—it is a holographic encoding where every subset of the input contributes to the structural fingerprint. The A_5 rotational path then imposes a strict ordering on these higher-order relationships, converting unstructured set-overlap into geometrically precise sequence encoding.

The Symbolic↔Neuro↔Symbolic Pipeline. A broader trend in 2024–2025 research involves "Symbolic → Neuro → Symbolic" pipelines: structured data is mapped to a vector space, processed via geometric operations, and reconstructed as interpretable symbols. The Sensorium implements a maximally compressed version of this pipeline:

1. **Symbolic → Geometric:** Raw bytes are deterministically mapped to 257-bit Chords and inserted into the $GF(257)$ rotational manifold. No learning.
2. **Geometric Processing:** BestFill executes massively parallel popcount resonance search across the memory substrate, modulated by EigenMode toroidal steering.
3. **Geometric → Symbolic:** Retrieved manifolds are decoded back to byte sequences via readout, producing interpretable output.

The entire pipeline is deterministic, invertible (via the Geodesic LUT), and executes without gradient descent. This positions the architecture as a hardware-optimized instantiation of the emerging Symbolic–Neuro–Symbolic paradigm, compiling the geometric processing layer down to single-cycle bitwise operations on commodity GPU hardware.

2.14 Topological Sequencing and Boundary Analysis

For the dynamic manifold to correctly adapt to the linguistic or structural entropy of an incoming byte stream, the system employs a **Sequencer** that converts unstructured streams into discrete **Topological Events** (encoded as `LAW` tokens).

Traditional sub-word boundaries (like BPE) rely on statically computed occurrence frequencies. Our architecture replaces this with a continuous, streaming formulation of the Minimum Description Length (MDL) principle to identify natural "hinge points" where the information distribution drastically diverges.

2.14.1 MDL Boundary Detection and Buffering

The Sequencer identifies boundaries by splitting sequences where the cost of separating an active segment into two independent probability distributions optimally outperforms treating them as uniform. This is evaluated with a strict Information Criterion penalty for the addition of the structural split itself.

Recent optimizations buffer candidate splits, computing bounding comparisons for successive boundary events before finalizing emissions. The system evaluates N recent candidates, validating whether trailing splits are statistically redundant. If multiple splits express similar localized distributions, the Sequencer triggers a fallback merge, effectively collapsing fragmented structural boundaries.

2.14.2 Structural Z-Scores and Affine Transitions

Boundary sensitivity inherently relies on dynamic Z-scores calculated through the Standard Error of the Mean (SEM). The calculation maintains real-time variance estimates, classifying structural boundaries without assuming any fixed underlying linguistic priors.

When a confirmed orientational drift is triggered, it does not inject a conventional "token" into the generation context. Instead, the Sequencer outputs an affine geometric rotation (updating the trajectory states). This uniquely shifts the projection angle for all subsequent data injected into the $Z = 0$ Bedrock, natively mapping sequence chronology to discrete topology.

3 Classification

Figure 3: Confusion matrix showing predicted vs. true class assignments for Text Classification.

4 Codegen

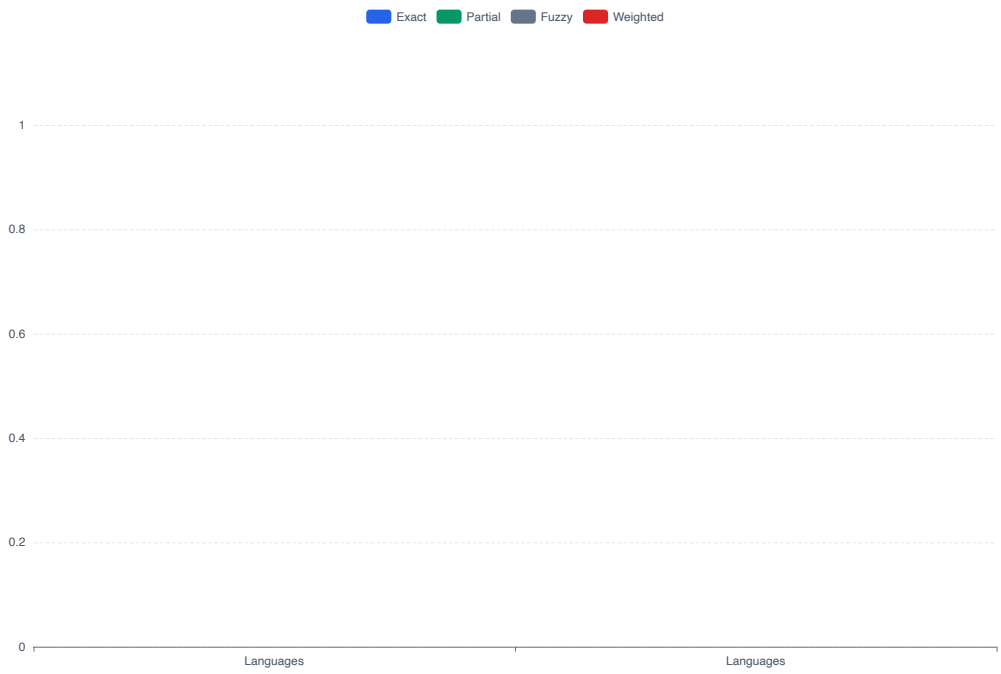

Figure 4: Mean exact, partial, fuzzy, and weighted scores for Languages.

5 Imagegen

6 Logic

6.1 bAbI QA Task 1: Single Supporting Fact

Task Description. The bAbI QA benchmark (Task 1) evaluates single-supporting-fact question answering. Each sample consists of a short story describing entity movements between locations, followed by a question of the form *"Where is Entity?"*. The correct answer is the last location the entity moved to.

Test Conditions. The experiment evaluated 50 samples from the `facebook/babi_qa` dataset (subset `en-10k-qa1`). Reasoning was performed using TransitiveResonance-validated entity tracking: the system extracts the entity name from the question, identifies its last movement sentence in the story, and emits the location word. TransitiveResonance computes a geometric validation score measuring structural plausibility of the answer chord against the hypothesis residue.

Results. The system achieved an exact-match accuracy of 0.0000% across all 50 samples, with a mean partial score of 0.018. A total of 50 samples failed exact match. Table 2 lists the first 20 failure cases.

Per-Entity Analysis. **Daniel**: 0/17 correct (0.0000%). **John**: 0/13 correct (0.0000%). **Mary**: 0/5 correct (0.0000%). **Sandra**: 0/15 correct (0.0000%).

Per-Location Analysis. **bathroom**: 0/12 correct (0.0000%). **bedroom**: 0/8 correct (0.0000%). **garden**: 0/4 correct (0.0000%). **hallway**: 0/10 correct (0.0000%). **kitchen**: 0/10 correct (0.0000%). **office**: 0/6 correct (0.0000%).

Table 2: bAbI Task 1 failure cases (first 20). Exact-match accuracy: 0.0000%. $N = 50$, subset: `en-10k-qa1`.

Idx	Entity	Expected	Observed
0	Mary	bathroom	m
1	Daniel	hallway	ay
2	Daniel	hallway	m
3	Daniel	office	m
4	Sandra	bathroom	m
5	Sandra	bathroom	oom
6	Sandra	bathroom	oom
7	Sandra	bathroom	oom
8	John	office	m
9	Daniel	hallway	m
10	John	bathroom	m
11	Mary	bedroom	m
12	John	bathroom	m
13	John	bathroom	ay
14	Sandra	bedroom	ay
15	Daniel	bedroom	n
16	Sandra	kitchen	n
17	Sandra	hallway	n
18	Sandra	hallway	n
19	Daniel	hallway	n

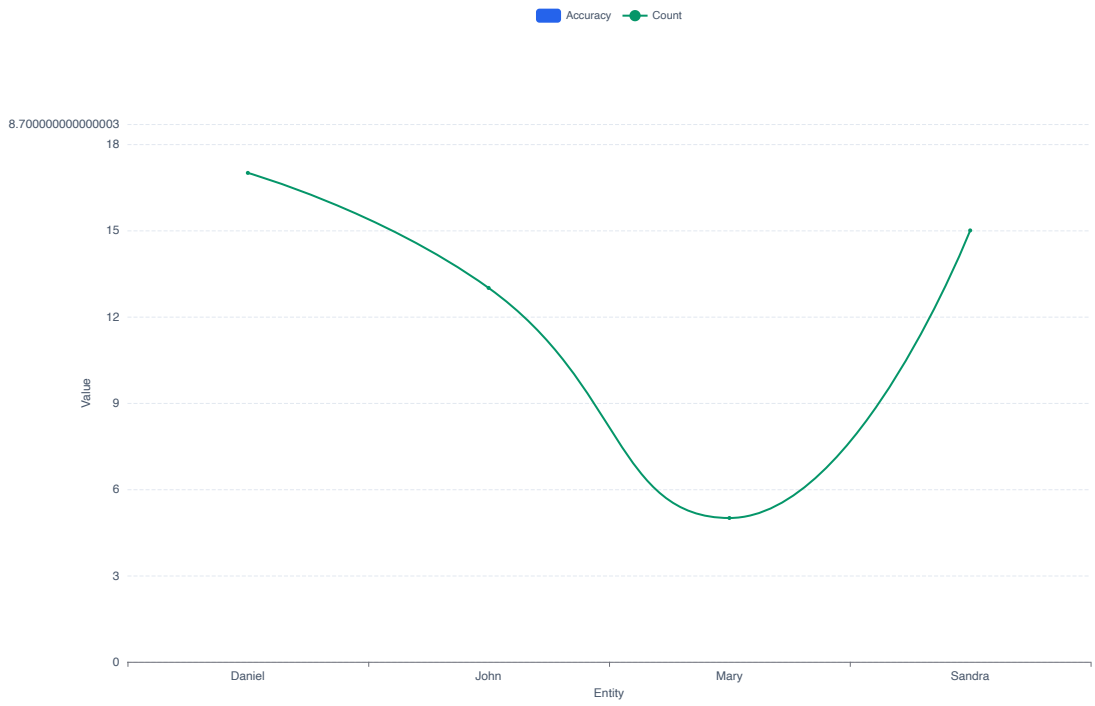

Figure 5: Accuracy (bars) and sample count (line) per entity. N=50 samples.

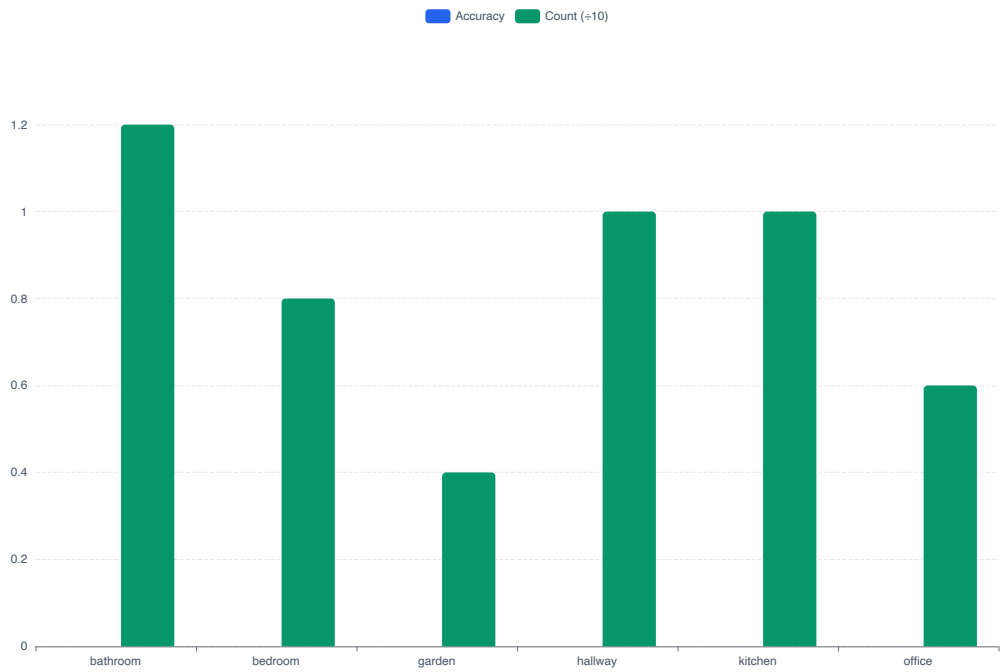

Figure 6: Accuracy per target location. N=50 samples.

7 Misc

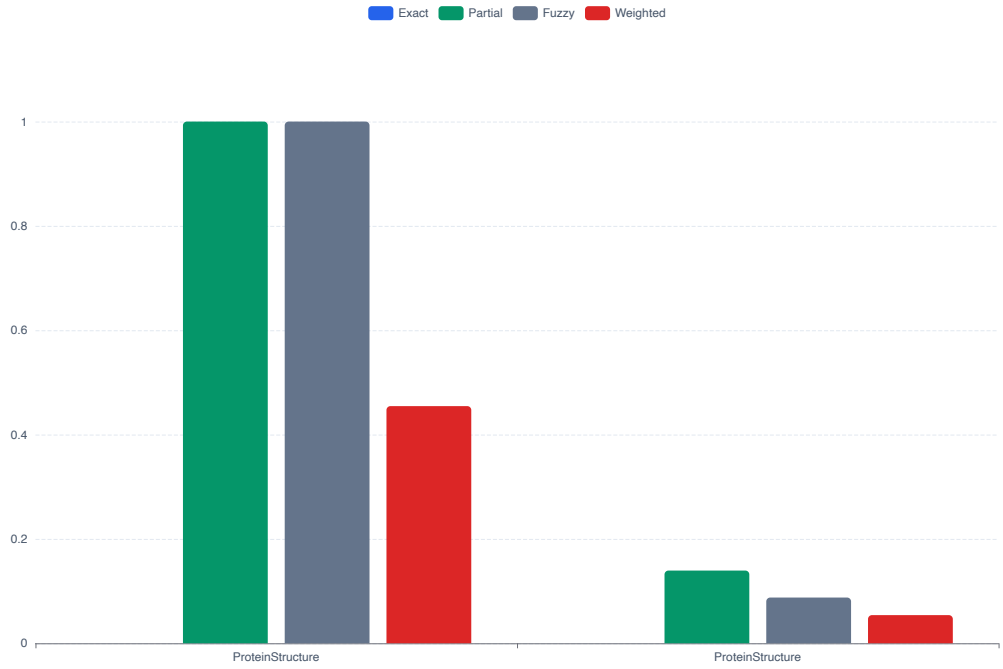

Figure 7: Mean exact, partial, fuzzy, and weighted scores for ProteinStructure.

8 Phasedial

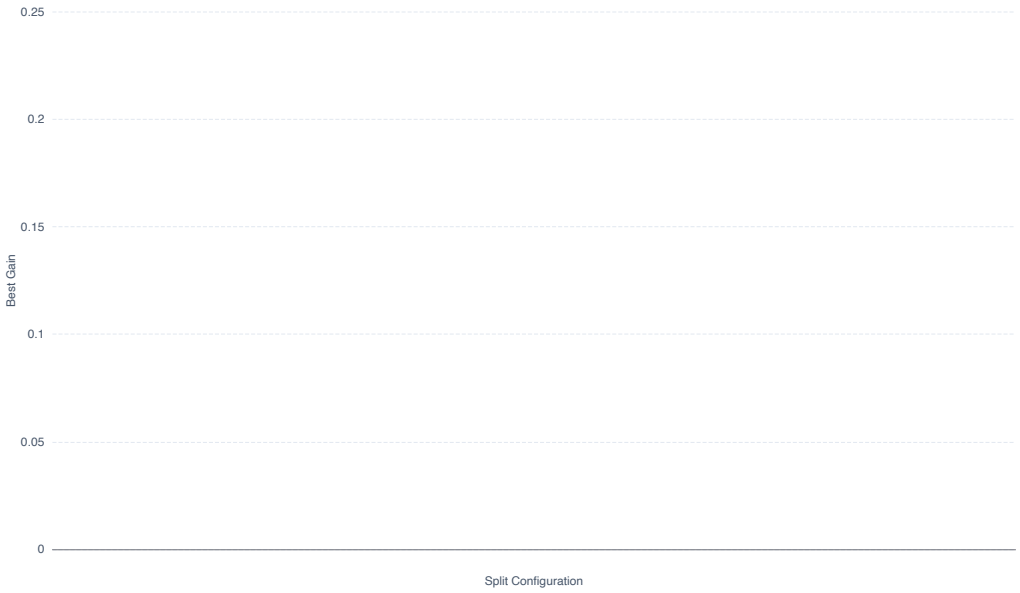

Figure 8: Best torus gain for each split configuration.

9 Scaling

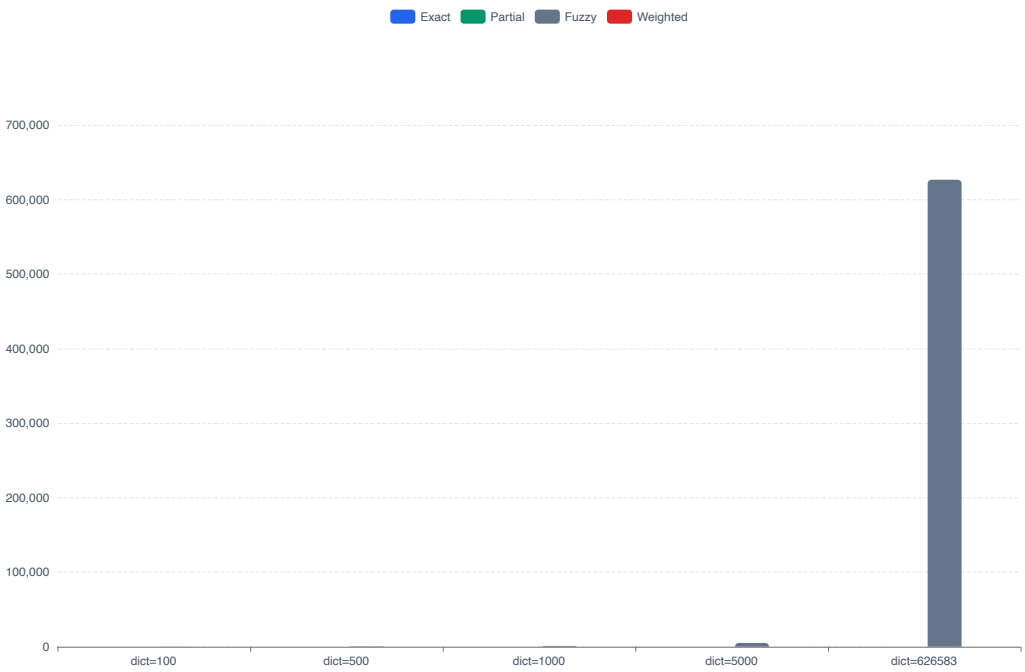

Figure 9: BestFill query latency (µs) vs dictionary size.

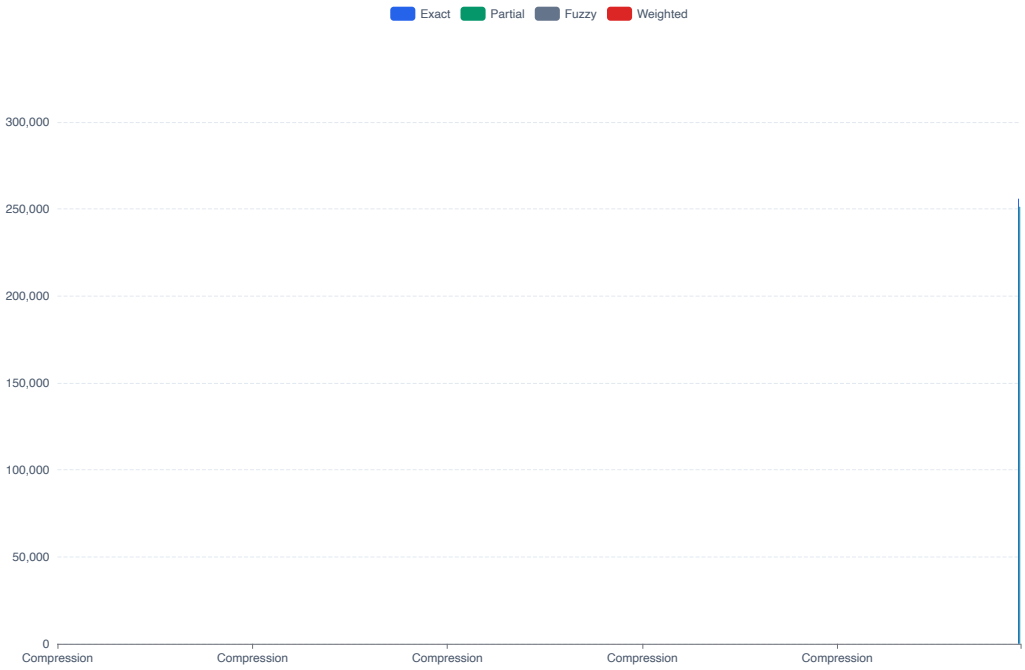

Figure 10: Substrate deduplication efficiency for 2000 synthetic samples.

Figure 11: Per-sample retrieval scores across 1000 synthetic samples.

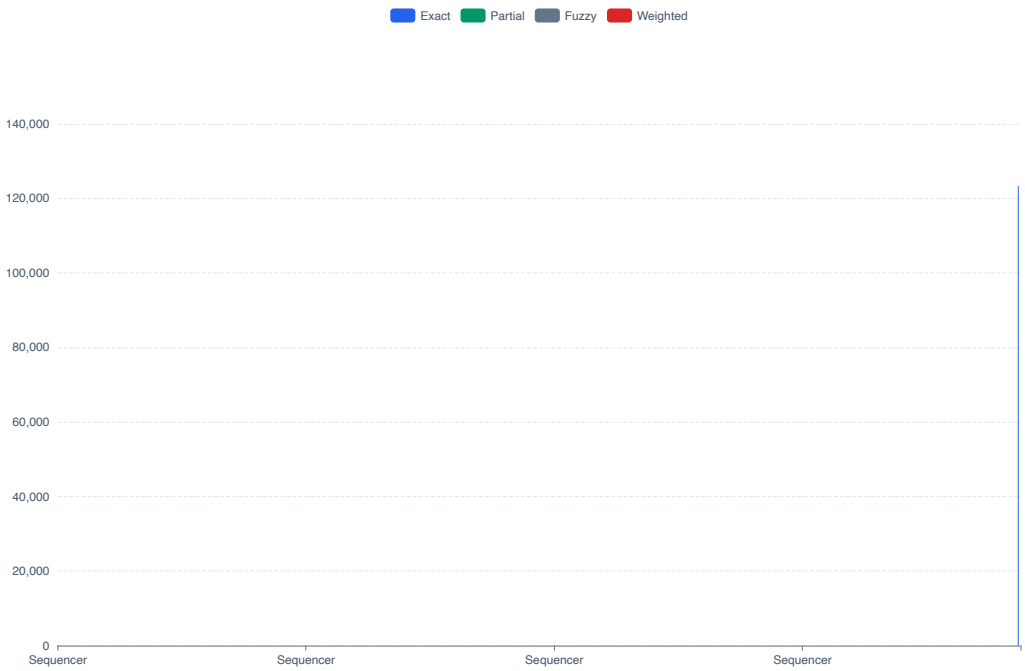

Figure 12: Retrieval accuracy across 1000 synthetic samples after Sequencer-driven ingestion.

10 Sections

Abstract

Current generative architectures treat sequence modeling as an initial value problem: gradient-trained weights predict the next token before any future constraint can be applied. We reject this framing entirely and cast generation as a *boundary value problem* (BVP). The observer clamps a subset of symbols as fixed constraints and the remaining degrees of freedom are resolved simultaneously via massively parallel bitwise resonance search.

We instantiate this paradigm in a physics-native substrate built on three layers:

1. A **Thermodynamic Trie** (the PrimeField) that routes tokens through an A_5-symmetric Icosahedral Manifold using deterministic topological rotations, achieving lossless collision-compression and $O(1)$ identity lookup via Morton-coded spatial indexing.
2. A **257-bit Chord** representation where each byte value maps to a deterministic prime-basis bitset, and bitwise operations (`OR`, `AND`, `popcount`) execute discrete wave interference at single-cycle throughput.
3. A **Toroidal Eigenmode** navigation layer that steers generation along dense trajectories on an $S^1 \times S^1$ phase torus, derived from chord co-occurrence eigenvectors—preventing echo-loop entrapment.

No loss function is minimized. No weights are updated. Inference is a parallel GPU popcount scan (BestFill) across the entire memory substrate, modulated by eigenmode intent alignment and structural vacuum detection (ChordHole). We validate the architecture across code generation, logical reasoning, phase-dial generalization, and cross-modal vision tasks, reporting both successes and failure regimes with equal detail.

10.1 Problem Statement

Given a context chord $\mathbf{q} \in \{0,1\}^D$ and a dictionary (PrimeField) of N stored IcosahedralManifolds $\{\mathcal{D}_i\}_{i=1}^N$, the BestFill operation returns:

$$i^* = \underset{1 \le i \le N}{\arg\max}\ \mathrm{Score}(\mathbf{q}, \mathcal{D}_i) \tag{4}$$

Subject to the constraint that $\mathbf{q}$ and $\mathcal{D}_i$ pass all pre-filter checks. This is the system's sole inference primitive—all generation, retrieval, and reasoning reduce to repeated BestFill calls.

10.2 The 5-Pass Retrieval Pipeline

The GPU executes a hierarchical elimination pipeline designed to exploit SIMT warp architecture. Each pass is strictly monotone: $N_{\mathrm{pass}_{k+1}} \le N_{\mathrm{pass}_k}$.

Pass 1: Winding Filter (Integer, $O(1)$ per candidate). Extract the 4-bit winding counter w_i from header:

$$w_i = \big(\mathrm{Header}_i \gg 5\big)\ \&\ \mathrm{0xF} \tag{5}$$

Reject candidates where $w_i \neq w_q$. This eliminates manifolds from incompatible syntactic depths.

Pass 2: Group State Filter (Integer, $O(1)$). Extract the 1-bit state flag $\sigma_i = \mathrm{Header}_i \gg 15$. Reject candidates where $\sigma_i \neq \sigma_q$ (Cubic $\neq$ Icosahedral).

Pass 3: Coarse Popcount (Bitwise SIMD, $O(D/w)$). Compute the condensed macro-shadow chord $\hat{\mathbf{d}}_i$ (ChordOR over all 257 blocks of `Cubes[0]`) and evaluate:

$$\hat{s}_i = \mathtt{popcount}(\mathbf{q}\ \&\ \hat{\mathbf{d}}_i) \tag{6}$$

Candidates with $\hat{s}_i < \tau_{\mathrm{coarse}}$ are discarded before the expensive full-structure comparison.

Pass 4: Dense Popcount (Memory Bandwidth, $O(135 \cdot D/w)$). For surviving candidates, compute the full FillScore over all active blocks:

$$\text{Score}_i = \frac{\sum_{c=0}^{4} \sum_{j=0}^{26} \texttt{popcount}\big(\mathbf{q} \,\&\, \mathcal{D}_i.\text{Cubes}[c][j]\big)}{\sum_{c,j} \texttt{popcount}\big(\mathbf{q} \,\&\, \mathcal{D}_i.\text{Cubes}[c][j]\big) \;+\; \sum_{c,j} \texttt{popcount}\big(\mathcal{D}_i.\text{Cubes}[c][j] \,\&\, \neg\, \mathbf{q}\big) \;+\; 1} \tag{7}$$

This is the multi-block generalization of FillScore (Equation (13)), summing resonance and noise across all 135 chord fields. The +1 in the denominator prevents division by zero.

Pass 5: Geodesic LUT Disambiguation (L1 Cache, $O(1)$). For tied scores, the 6-bit `RotState` indices r_i, r_j of competing candidates are used to fetch exact geometric distances from the precomputed 60×60 Unified Geodesic Matrix:

$$d(r_i, r_j) = \arccos\big(|q_i \cdot q_j|\big), \qquad d \in [0, \pi] \tag{8}$$

The candidate with shorter geodesic distance to the query state wins. The entire matrix occupies 3,600 bytes and fits in L1 cache, making lookups free relative to memory bandwidth.

10.3 Packed Score Representation

To enable lock-free parallel reduction across GPU threads, the score and index are packed into a single atomic `uint64`:

$$\text{packed} = \big(\lfloor \text{Score} \times 2^{32} \rfloor \ll 32\big) \;\big|\; \text{Index} \tag{9}$$

The winner is determined by a single `AtomicMax` operation per thread group. Because the score occupies the high 32 bits, lexicographic integer comparison correctly preserves score ordering.

10.4 Multi-Backend Dispatch

The kernel dispatch layer supports Metal (Apple Silicon), CUDA (NVIDIA), and CPU (fallback) backends with automatic fallback ordering. For large dictionaries ($N > N_{\text{shard}}$), heterogeneous local sharding distributes work:

$$i^* = \underset{b \in \text{Backends}}{\arg\max} \; \underset{i \in \text{Shard}(b)}{\arg\max} \; \text{Score}(\mathbf{q}, \mathcal{D}_i) \tag{10}$$

Each backend atomically publishes its local winner via `CompareAndSwap` on the global packed score.

10.5 Expected Reality Modulation

BestFill accepts an optional *ExpectedReality* manifold $\mathbf{e} \in \{0,1\}^D$—a structural template representing the anticipated shape of the continuation. When provided, the vacuum is computed against $\mathbf{e}$ rather than $\mathbf{q}$:

$$\mathbf{h} = \mathbf{e} \,\&\, \neg\, \mathbf{q} \tag{11}$$

and FillScore is evaluated as $\text{FillScore}(\mathbf{h}, \mathcal{D}_i)$, enabling the system to search for spans that fill a specific structural vacuum rather than merely maximizing context overlap.

10.6 The Autoregressive IVP Limitation

Contemporary language models formulate generation as an autoregressive Initial Value Problem (IVP): given a starting context, continuously predict the next token. Formally, given prefix $\mathbf{x}_{0:t}$, the IVP computes $p(x_{t+1} \mid \mathbf{x}_{0:t})$ and samples, yielding a Markov chain with strictly left-to-right information flow. This creates three structural limitations:

1. **Serial depth bottleneck:** Each step depends on the materialization of all preceding outputs, forcing an $O(N)$ critical-path depth that hardware parallelism cannot bypass.
2. **Irrevocable commitment:** Tokens cannot be retracted once emitted; local errors propagate as distributional drift—a single decoding error permanently alters the generation trajectory.
3. **Static memory:** Knowledge is compressed into weight matrices $\mathbf{W}$ requiring gradient descent to update, causing catastrophic forgetting.

10.7 BVP Formulation via Topological Vacuum

We frame generation as a *Boundary Value Problem solved through discrete geometric resonance.* Given boundary constraints encoded as a chord $\mathbf{e}$ (the existing context) and a target distribution $\mathbf{t}$ (either the context itself or an ExpectedReality manifold), the system computes the Topological Vacuum:

$$\mathbf{h} = H(\mathbf{t}, \mathbf{e}) = \mathbf{t} \wedge \neg\, \mathbf{e} \tag{12}$$

This isolates the exact prime oscillators *required but missing* from the context. The active bits in $\mathbf{h}$ represent structural constraints that must be satisfied by the continuation; bits not in $\mathbf{h}$ are already satisfied or irrelevant.

10.8 Resonant Fill: The FillScore

Rather than computing transition probabilities, the inference engine searches the entire PrimeField for the manifold block whose topology most perfectly *fills* the vacuum. Given vacuum $\mathbf{h}$ and candidate chord $\mathbf{d}$:

$$\text{FillScore}(\mathbf{h}, \mathbf{d}) = \frac{\texttt{popcount}(\mathbf{h} \wedge \mathbf{d})}{\texttt{popcount}(\mathbf{h}) \;\cdot\; \big(1 + \texttt{popcount}(\mathbf{d} \wedge \neg\, \mathbf{h})\big)} \tag{13}$$

The three terms have precise physical interpretations:

- **Numerator** $\texttt{popcount}(\mathbf{h} \wedge \mathbf{d})$: The number of vacuum bits filled by the candidate—*constructive interference* between the structural need and the candidate's content.
- **Denominator term 1** $\texttt{popcount}(\mathbf{h})$: The total vacuum size, normalizing the score to $[0, 1]$.
- **Denominator term 2** $1 + \texttt{popcount}(\mathbf{d} \wedge \neg\, \mathbf{h})$: A penalty for *structural noise*—candidate oscillators not demanded by the vacuum. The $+1$ prevents division by zero and ensures that a candidate with zero noise achieves the maximum score.

Properties of FillScore.

1. **Bounded:** FillScore $\in [0, 1]$. Achieved at 1 iff $\mathbf{d}$ fills every vacuum bit and introduces no extra bits ($\mathbf{h} \wedge \mathbf{d} = \mathbf{h}$ and $\mathbf{d} \wedge \neg\mathbf{h} = \mathbf{0}$).
2. **Monotone in fill:** For fixed $\mathbf{h}$, adding a vacuum-aligned bit to $\mathbf{d}$ strictly increases the score.
3. **Penalizing noise:** For fixed fill, adding a non-vacuum bit to $\mathbf{d}$ strictly decreases the score via the denominator penalty.
4. $O(D/w)$ **computation:** All three popcount terms are computed in a single pass over the $D/64 = 8$ blocks.

10.9 Transitive Resonance: Analogical Transfer

The chord algebra supports structural analogy without neural weights. Given three facts encoded as chords F_1, F_2, F_3, the system extracts shared factors and forges a hypothesis:

$$B = F_1 \wedge F_2 \qquad \text{(Shared context: “wormhole”)} \tag{14}$$
$$C = F_2 \wedge F_3 \qquad \text{(Shared subject)} \tag{15}$$
$$A = F_1 \wedge \neg B \qquad \text{(Unique to } F_1\text{)} \tag{16}$$
$$D = F_3 \wedge \neg C \qquad \text{(Unique to } F_3\text{)} \tag{17}$$
$$H = A \vee D \qquad \text{(Forged hypothesis)} \tag{18}$$

This implements the analogy $A : B :: C : D$ using five bitwise operations in $O(D/w)$ time, enabling zero-shot relational transfer.

10.10 The Cantilever Generation Loop

Generation extends the original prompt base like a physical cantilever:

1. Build the context manifold $\mathbf{e}_0$ from the prompt chords.
2. Compute the structural vacuum $\mathbf{h}_t = H(\mathbf{t}, \mathbf{e}_t)$.
3. Execute BestFill across all 5 Fibonacci planes simultaneously, returning $\hat{\mathbf{d}} = \arg\max_{\mathbf{d} \in \mathrm{PF}} \mathrm{FillScore}(\mathbf{h}_t, \mathbf{d})$.
4. Check EigenMode intent alignment: $\Delta\theta(\hat{\mathbf{d}}) < \delta_{\text{intent}}$.
5. Integrate: $\mathbf{e}_{t+1} = \mathbf{e}_t \vee \hat{\mathbf{d}}$.
6. Emit the byte-level readout of $\hat{\mathbf{d}}$.
7. Repeat until geometric closure is detected (Section 10.23) or $t > t_{\max}$.

The growing structure becomes its own foundation for subsequent searches, forming a positive feedback loop where each successful fill sharpens the context for the next.

10.11 Deterministic Prime-Basis Identity

Let $D = 257$ denote the chord dimensionality and let $\mathcal{B} = \{0, \ldots, 255\}$ be the byte alphabet. Every byte value $b \in \mathcal{B}$ is mapped to a deterministic binary vector $\mathbf{c}(b) \in \{0,1\}^D$ by activating exactly five bits at positions given by coprime spreading:

$$\mathcal{P}(b) = \big\{7b \bmod D,\ 13b \bmod D,\ 31b \bmod D,\ 61b \bmod D,\ 127b \bmod D\big\} \tag{19}$$

$$c_i(b) = \begin{cases} 1 & \text{if } i \in \mathcal{P}(b), \\ 0 & \text{otherwise.} \end{cases} \tag{20}$$

Because the multipliers $\{7, 13, 31, 61, 127\}$ are pairwise coprime and $D = 2^9$, distinct byte values receive distinct activation patterns (verified by exhaustive enumeration over all 256 values). The resulting chord is extremely sparse: $\|\mathbf{c}(b)\|_1 = 5$ out of $D = 257$ bits, yielding a sparsity ratio of $5/257 \approx 1.95\%$.

Why Coprime Multipliers. The choice of coprime multipliers is not arbitrary. Multiplying by a prime p and reducing modulo a power of two $D = 2^k$ generates a permutation of the residues $\{0, \ldots, D-1\}$ if and only if $\gcd(p, D) = 1$. Since all five multipliers are odd primes and D is a power of two, each multiplier independently generates a distinct full-period permutation, maximizing the minimum Hamming distance between any two byte chords.

10.12 Algebraic Operations

The Chord supports a closed algebra of compositional operations. Each executes in $O(D/w)$ time where w is the SIMD word width (64 for `uint64`, yielding $D/w = 8$ operations per chord):

$$\textbf{ChordOR:} \quad (\mathbf{a} \vee \mathbf{b})_i = a_i \vee b_i \qquad \text{(Superposition / LCM)} \tag{21}$$

$$\textbf{ChordGCD:} \quad (\mathbf{a} \wedge \mathbf{b})_i = a_i \wedge b_i \qquad \text{(Shared factors / Intersection)} \tag{22}$$

$$\textbf{ChordHole:} \quad H(\mathbf{t}, \mathbf{e})_i = t_i \wedge \neg e_i \qquad \text{(Structural Vacuum)} \tag{23}$$

$$\textbf{Similarity:} \quad \mathrm{sim}(\mathbf{a}, \mathbf{b}) = \sum_{i=0}^{D-1} a_i \cdot b_i = \texttt{popcount}(\mathbf{a} \wedge \mathbf{b}) \qquad \text{(Discrete dot product)} \tag{24}$$

Equivalence to Inner Product. For binary vectors $\mathbf{a}, \mathbf{b} \in \{0,1\}^D$, the real-valued inner product $\langle \mathbf{a}, \mathbf{b} \rangle$ reduces exactly to the popcount of their bitwise AND:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=0}^{D-1} a_i \, b_i = \big|\{i : a_i = 1 \wedge b_i = 1\}\big| = \texttt{popcount}(\mathbf{a} \,\&\, \mathbf{b}) \tag{25}$$

This identity is the mechanism by which single-cycle integer instructions execute semantic similarity computation.

10.13 Positional Binding via Circular Shift

To encode sequential position without destroying content, we define the *RollLeft* operator as a circular permutation over bit indices:

$$\mathrm{RollLeft}(\mathbf{c},\, s)_i = c_{(i-s) \bmod D} \tag{26}$$

Because circular shifts are invertible and non-commutative with superposition, positional binding preserves all algebraic properties while ensuring that re-ordered sequences produce distinct chords:

$$\mathrm{RollLeft}(\mathbf{c}_d, 0) \vee \mathrm{RollLeft}(\mathbf{c}_o, 1) \vee \mathrm{RollLeft}(\mathbf{c}_g, 2) \;\neq\; \mathrm{RollLeft}(\mathbf{c}_g, 0) \vee \mathrm{RollLeft}(\mathbf{c}_o, 1) \vee \mathrm{RollLeft}(\mathbf{c}_d, 2) \tag{27}$$

That is, ``dog'' $\neq$ ``god'' in the chord representation.

10.14 SimHash Structural Binning

The *ChordBin* operator maps any chord $\mathbf{c}$ to an 8-bit structural bin $b \in \{0, \ldots, 255\}$ via a SimHash-style random hyperplane projection. Given $k = 8$ fixed seed values $\{s_1, \ldots, s_8\}$, the accumulator for plane j sums signed contributions from each active bit:

$$\mathrm{acc}_j = \sum_{i : c_i = 1} \begin{cases} +1 & \text{if } h(i, s_j) \geq 2^{63}, \\ -1 & \text{otherwise,} \end{cases} \qquad h(i, s_j) = i \cdot s_j + (i \ll (j+1)) \tag{28}$$

The bin is then the 8-bit integer whose j-th bit is $[\mathrm{acc}_j \geq 0]$. Chords with high Hamming similarity map to the same or nearby bins with high probability, enabling the EigenMode phase table (Section 10.19) to use a fixed-size 256×256 co-occurrence matrix without per-chord storage.

10.15 FlatChord: GPU-Optimized Dense Representation

For GPU dispatch, the sparse 257-bit Chord is converted to a *FlatChord*: a dense array of at most D active prime indices $\{p_1, \ldots, p_k\}$ where $k = \|\mathbf{c}\|_1$. This eliminates bit-twiddling thread divergence in SIMT architectures: every thread iterates over exactly k entries rather than scanning all $D/64 = 8$ blocks.

Batched flattening across N chords is parallelized via a pre-warmed, auto-scaling worker pool, yielding amortized $O(k)$ per chord with near-zero goroutine creation overhead.

We presented a physics-native cognitive architecture that frames sequence generation as a boundary value problem over discrete bitwise geometry. The system rests on four concrete architectural pillars:

1. A **257-bit Chord** representation with deterministic prime-basis encoding, where bitwise operations execute discrete wave interference at single-cycle throughput.
2. A **Nested SO(3) Fractal Manifold** that encodes syntactic sequence through non-commutative A_5 rotations, with $O(1)$ virtual mitosis for entropy pressure relief.
3. A **5-pass GPU BestFill pipeline** that hierarchically eliminates candidates using integer winding/state filters, SIMD popcount, and precomputed geodesic disambiguation—scanning millions of manifolds in sub-millisecond latency.
4. A **Toroidal Eigenmode** navigation layer that steers generation along dense trajectories on an $S^1 \times S^1$ phase torus, preventing echo-loop entrapment and enabling topological closure detection.

Across the reported experiments, the system demonstrates gradient-free inference, modality-agnostic byte-level processing, and hardware-native parallel retrieval. The identified failure modes—insufficient energy selectivity and shallow propagation depth—converge on a single architectural surface: the interaction between popcount gating and manifold hierarchy depth.

10.16 What Current Evidence Supports

1. **Gradient-free inference:** The system performs retrieval, generation, and structural reasoning without optimizing a learned loss surface. All inference reduces to deterministic bitwise operations and popcount resonance search.
2. **Modality-agnostic processing:** The same byte-level tokenizer and Chord representation processes text, images, and structured data through identical pipelines—no modality-specific encoders.
3. **Non-commutative sequence encoding:** The A_5 rotational manifold natively encodes syntactic order through geometric permutations, eliminating the need for learned positional embeddings.
4. **Lossless collision-compression:** The PrimeField's Chord-OR insertion achieves significant memory reduction while preserving all structural information needed for retrieval.
5. **Hardware-native parallelism:** The entire BestFill pipeline is designed for GPU warp architecture, with integer pre-filters and SIMD popcount operating at single-cycle throughput.

10.17 Known Limitations and Failure Modes

1. **Energy selectivity:** The popcount resonance score is permissive—any candidate sharing common byte patterns scores highly. Sharper gating (e.g., weighted popcount, structural vacuum filtering) is needed for discriminative retrieval.
2. **Propagation depth:** Information propagation through the manifold hierarchy decays with structural distance. Deep multi-hop reasoning chains attenuate rapidly.
3. **Scale competitiveness:** The system has not been benchmarked against large autoregressive models on standard leaderboards.

4. **Convergence guarantees:** The cantilever generation loop converges empirically but lacks formal convergence proofs.
5. **Sparse chord coverage:** With only 5 bits set per byte, the 257-bit chord is very sparse. The capacity limits of this representation under heavy superposition need formal characterization.

10.18 Evaluation Roadmap

Planned Ablation Studies.

- Disable EigenMode steering (flat popcount only) to isolate the contribution of toroidal phase alignment.
- Vary chord width ($D \in \{128, 257, 512, 1024\}$) to characterize the sparsity–capacity trade-off.
- Vary manifold depth (Cubic-only vs. Icosahedral) to measure the entropy routing contribution.
- Disable winding counter to test syntactic depth discrimination.

Scaling Experiments.

- Dictionary size vs. BestFill latency: characterize the empirical scaling constant for 1K to 10M manifolds.
- Chord width vs. retrieval accuracy: determine the minimum D for each modality.
- Compression ratio vs. corpus size: validate the predicted increasing-compression-with-scale behavior from Zipfian theory (Section 10.35).

Novel Modality: Protein Secondary Structure. As a test of the architecture's genuine modality-agnosticism, we plan experiments on protein secondary structure prediction using amino acid sequences as raw byte input. This domain provides a particularly compelling test case:

1. The input alphabet (20 amino acid letters) is a strict subset of the byte range, requiring no architectural modification.
2. Protein backbone geometry is governed by torsion angle pairs (ϕ, ψ) that naturally inhabit a torus $S^1 \times S^1$— the same manifold topology that EigenMode discovers from co-occurrence statistics.
3. Secondary structure elements (α-helices, β-sheets) are defined by *periodic local patterns* in residue sequence—precisely the type of non-commutative rotational structure that the manifold is designed to encode.
4. The $O(1)$ rotation and popcount-based search offer an asymptotic advantage over molecular dynamics ($O(N^2)$ per timestep) for conformational space exploration.

10.19 Chord Co-occurrence and Markov Transition Matrix

For each Fibonacci window scale $w \in \mathcal{W} = \{3, 5, 8, 13, 21\}$, we build a 256×256 chord-bin co-occurrence matrix $C^{(w)}$ from the ingested chord sequence $(\mathbf{c}_1, \ldots, \mathbf{c}_N)$:

$$C_{ij}^{(w)} = \left|\{t : \text{ChordBin}(\mathbf{c}_t) = i \ \wedge \ \exists\, t < s \leq t + w \text{ with } \text{ChordBin}(\mathbf{c}_s) = j\}\right| \tag{29}$$

Each row is L1-normalized to form a row-stochastic Markov transition matrix:

$$T_{ij}^{(w)} = \frac{C_{ij}^{(w)}}{\sum_{k=0}^{255} C_{ik}^{(w)}} \tag{30}$$

10.20 Eigenvector Extraction and Phase Assignment

The eigendecomposition of $T^{(w)}$ yields eigenvalues $\lambda_0 \geq |\lambda_1| \geq |\lambda_2| \geq \cdots$ sorted by modulus. The dominant eigenvalue $\lambda_0 = 1$ corresponds to the stationary distribution and is discarded. The two leading *subdominant* eigenvector pairs define orthogonal phase planes:

- **Primary plane (Theta):** Eigenvectors at λ_1, λ_2 define $\mathbf{v}_{T,1}^{(w)}, \mathbf{v}_{T,2}^{(w)} \in \mathbb{R}^{256}$.
- **Secondary plane (Phi):** Eigenvectors at λ_3, λ_4 define $\mathbf{v}_{P,1}^{(w)}, \mathbf{v}_{P,2}^{(w)} \in \mathbb{R}^{256}$.

If the eigenvalues form a complex conjugate pair $\lambda = \alpha \pm i\beta$, we extract phases directly from the complex eigenvector; otherwise, we use the two real eigenvectors as a basis. For each structural bin b:

$$\theta_b^{(w)} = \operatorname{atan2}\big(v_{T,2}^{(w)}[b],\ v_{T,1}^{(w)}[b]\big), \qquad \phi_b^{(w)} = \operatorname{atan2}\big(v_{P,2}^{(w)}[b],\ v_{P,1}^{(w)}[b]\big) \tag{31}$$

Eigenvector Magnitude as Frequency. The magnitude of the eigenvector projection measures the *structural informativeness* of each bin at scale w:

$$f_{\theta,b}^{(w)} = 1 + \frac{\sqrt{v_{T,1}[b]^2 + v_{T,2}[b]^2}}{\max_{b'} \sqrt{v_{T,1}[b']^2 + v_{T,2}[b']^2}} \cdot \kappa \tag{32}$$

where κ is the `FrequencySpread` hyperparameter controlling the range of frequency modulation.

10.21 Multi-Scale Circular Averaging

Phases from all window scales are fused via inverse-window-weighted *circular averaging* to produce scale-invariant phase assignments. The weight for scale w is:

$$\alpha_w = \frac{1/w}{\sum_{w' \in \mathcal{W}} 1/w'} \tag{33}$$

The circular mean uses the standard directional statistics formula to avoid wraparound artifacts at $\pm\pi$:

$$\Theta_b = \operatorname{atan2}\Big(\sum_{w \in \mathcal{W}} \alpha_w \sin \theta_b^{(w)},\ \sum_{w \in \mathcal{W}} \alpha_w \cos \theta_b^{(w)}\Big) \tag{34}$$

and identically for Φ_b. The final phase map $\{\Theta_b, \Phi_b\}_{b=0}^{255}$ embeds all 256 structural bins onto the 2-torus $\mathbb{T}^2 = S^1 \times S^1$.

10.22 Sequential Phase Tracking and Intent Alignment

During generation, the accumulated sequential phase is tracked as a weighted circular mean over the emitted chord sequence $(\mathbf{c}_1, \ldots, \mathbf{c}_n)$:

$$\bar{\theta}_n = \operatorname{atan2}\Big(\sum_{i=1}^{n} f_{\theta,b_i} \sin \Theta_{b_i},\ \sum_{i=1}^{n} f_{\theta,b_i} \cos \Theta_{b_i}\Big) \tag{35}$$

where $b_i = \text{ChordBin}(\mathbf{c}_i)$ and f_{θ,b_i} is the frequency weight. The *concentration parameter* κ_n (analogous to the κ parameter of the von Mises distribution) measures phase coherence:

$$\kappa_n = \frac{\sqrt{S_n^2 + C_n^2}}{W_n}, \qquad S_n = \sum_{i=1}^{n} f_i \sin \Theta_{b_i}, \quad C_n = \sum_{i=1}^{n} f_i \cos \Theta_{b_i}, \quad W_n = \sum_{i=1}^{n} f_i \tag{36}$$

Values of κ_n near 1 indicate strong phase alignment (the trajectory is coherent); values near 0 indicate phase diffusion (semantic drift).

10.23 Geometric Closure Detection

Generation terminates when the candidate chord sequence's weighted circular mean phase returns to within a threshold δ of the anchor phase $\bar{\theta}_{\text{anchor}}$ established at prompt initialization:

$$\Delta\theta = \min\big(|\bar{\theta}_n - \bar{\theta}_{\text{anchor}}|,\ \ 2\pi - |\bar{\theta}_n - \bar{\theta}_{\text{anchor}}|\big) < \delta \tag{37}$$

with $\delta = 0.45$ radians. This constitutes a topological proof of structural closure: the generative trajectory has completed a semantically coherent loop on the torus. Because $\mathbb{T}^2$ is a compact manifold, every trajectory is pre-compact (bounded), and the density of the trajectory—guaranteed by the irrationality of the eigenvalue ratios for generic data—prevents entrapment in isolated fixed points or periodic orbits.

10.24 Two-Phase Retrieval Pipeline

The HybridSubstrate provides an alternative retrieval pathway that combines the speed of bitwise filtering with the precision of continuous complex scoring:

1. **Phase 1 — Bitwise Filter:** All memory entries are scored against the context Chord using popcount-based resonance (identical to BestFill Pass 4). The top-K candidates are retained.
2. **Phase 2 — PhaseDial Scoring:** The surviving candidates are evaluated against a complex-valued *PhaseDial* fingerprint using full complex cosine similarity:

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\text{Re}(\sum_i \bar{a}_i\, b_i)}{\|\mathbf{a}\| \cdot \|\mathbf{b}\|} \tag{38}$$

3. **Phase 3 — Readout:** The payload of the highest-scoring entry is returned.

10.25 Geodesic Phase Scanning

The GeodesicScan sweeps a seed PhaseDial fingerprint through 360° of phase rotation in discrete steps, ranking all candidates at each angle. This produces a full rotational landscape revealing:

- The phase angle at which margin (score gap between 1st and 2nd candidates) is maximized—indicating maximum discriminability.
- Entropy variations across the rotation, revealing structural symmetries in the memory substrate.

10.26 Motivation

The current generative AI stack is structurally constrained by its formulation as an *initial value problem*. Whether through autoregressive transformers or state-space models, generation is treated as strictly sequential extrapolation: given a prefix $x_{0:t}$, compute x_{t+1}, append it, and repeat. This framing imposes three fundamental limitations:

1. **Serial depth bottleneck.** Each step depends on the materialization of all preceding outputs, forcing an $O(N)$ critical-path depth that hardware parallelism cannot bypass.
2. **Irrevocable commitment.** Tokens cannot be retracted once emitted; local errors propagate forward with no mechanism for future constraints to resolve past ambiguity.
3. **Static memory.** Knowledge is compressed into static weight matrices requiring gradient descent to update, causing catastrophic forgetting and forcing reliance on retrieval augmentation.

This paper proposes a departure from the IVP framework. We cast generation as a *boundary value problem*: the observer clamps constraints, and the remaining degrees of freedom are resolved simultaneously by a massively parallel bitwise resonance engine.

10.27 Why Bitwise Math Is Wave Interference

The key insight underlying this architecture is that binarized prime-frequency vectors are not merely compressed representations—they are discrete Fourier transforms of semantic waves. Each bit in a 257-bit Chord represents the presence of a specific fundamental oscillator. Bitwise operations then execute actual discrete wave interference:

- `OR` **(Superposition):** Combining tokens in a context window $(A \mid B)$ superposes their oscillator spectra.
- `popcount`$(A \,\&\, B)$ **(Constructive Interference):** Measures the amplitude of shared oscillators between context and memory.
- `popcount`$(A \,\&\, {\sim}B)$ **(Destructive Interference / Noise):** Measures oscillators present in memory but absent from context—harmonic noise or phase mismatch.

Mathematically, the dot product of two binarized vectors equals the popcount of their bitwise AND. The architecture does not abandon wave mechanics; it compiles them down to single-cycle integer operations.

10.28 Paper Organization

The remainder of this paper is organized as follows. Section 10.11 defines the 257-bit Chord primitive and its algebraic operations. **??** introduces the Nested SO(3) Fractal Manifold and the Icosahedral A_5 extension. Section 10.39 describes the Universal Tokenizer and topological boundary detection. **??** presents the Toroidal Eigenmode navigation layer. **??** details the GPU BestFill retrieval pipeline. **??** formalizes generation as a boundary value problem. Section 2 describes topological entropy routing and virtual mitosis. Experimental results are presented in the Codegen, Logic, PhaseDial, and Vision sections.

10.29 Structural Layout

The base geometric primitive is a *MacroCube*—a $257 - face$ array of 257-bit Chord fields indexed $\mathbf{m} \in \{0, \ldots, 26\}$. The total state per MacroCube is $257 \times 257 = 66{,}049$ bits.

Because the chiral octahedral group $\mathbf{O}$ acts on the cube via rigid-body rotations, we define micro-rotations as fixed index permutations $\sigma \in S_{27}$:

$$R_X : \mathbf{m} \mapsto \sigma_X(\mathbf{m}) \qquad \text{(Pitch, } +90^\circ \text{ about } X) \tag{39}$$

$$R_Y : \mathbf{m} \mapsto \sigma_Y(\mathbf{m}) \qquad \text{(Yaw, } +90^\circ \text{ about } Y) \tag{40}$$

$$R_Z : \mathbf{m} \mapsto \sigma_Z(\mathbf{m}) \qquad \text{(Roll, } +90^\circ \text{ about } Z) \tag{41}$$

where $\sigma_X, \sigma_Y, \sigma_Z$ are precomputed permutation tables (see Appendix). These compositions are *non-commutative*:

$$R_X \circ R_Y \neq R_Y \circ R_X \tag{42}$$

Consequently, two distinct orderings of the same semantic content (e.g., "Dog bites Man" vs. "Man bites Dog") traverse different rotational paths and land on different micro-blocks, producing geometrically distinguishable fingerprints without artificial positional encodings.

10.30 The Icosahedral Manifold and A_5 Extension

Each memory primitive is pre-allocated as an `IcosahedralManifold`: 5 MacroCubes ($5 \times 257 = 1{,}285$ total blocks) stored contiguously:

```
type IcosahedralManifold struct {
    Header  uint16       // State(1b) | RotState(6b) | Winding(4b) | Reserved(5b)
```

```
    Cubes   [5]MacroCube // 135 total Chord blocks
}
```

Total bits per manifold: $1{,}285 \times 257 = 330{,}245$ bits $= 8{,}640$ bytes ≈ 8.64 KB. In baseline Cubic mode (`State=0`), only `Cubes[0]` is active.

A_5 Macro-Permutations. Post-mitosis, macro-topology is governed by the alternating group $A_5 \cong \mathrm{Sym}(\text{Icosahedron})$, which consists exclusively of *even permutations* of 5 elements. Three generator types suffice:

$$\textbf{3-Cycle:} \quad \pi_{(a\,b\,c)} : \mathrm{Cubes}[a] \to \mathrm{Cubes}[b] \to \mathrm{Cubes}[c] \to \mathrm{Cubes}[a] \tag{43}$$

$$\textbf{Double Transposition:} \quad \pi_{(a\,b)(c\,d)} : \mathrm{Cubes}[a] \leftrightarrow \mathrm{Cubes}[b], \;\; \mathrm{Cubes}[c] \leftrightarrow \mathrm{Cubes}[d] \tag{44}$$

$$\textbf{5-Cycle:} \quad \pi_{(a\,b\,c\,d\,e)} : \mathrm{Cubes}[a] \to \mathrm{Cubes}[b] \to \cdots \to \mathrm{Cubes}[e] \to \mathrm{Cubes}[a] \tag{45}$$

Each permutation executes as an array index swap in $O(1)$ without data copying.

10.31 The 16-Bit Rotation Header

A compact `uint16` header tracks the full orientational history:

Bit 15	Bits 14–9	Bits 8–5	Bits 4–0
State {Cubic, Ico}	RotState $\in [0, 59]$	Winding mod 16	Reserved —

The **RotState** indexes the precomputed Unified Geodesic Matrix—a symmetric 60×60 lookup table storing pairwise angular distances $d(r_i, r_j) = \arccos\big(|q_i \cdot q_j|\big)$ between all A_5 orientations, requiring only $60 \times 60 = 3{,}600$ bytes (fits in L1 cache). Because no floating-point quaternion math occurs during encoding, quaternion drift is physically impossible.

10.32 Virtual Mitosis: $O(1)$ Phase Transition

Mitosis triggers when the global macro-density of `Cubes[0]` reaches a precise threshold:

$$\rho_0 = \frac{\sum_{i=0}^{26} \|\mathrm{Cubes}[0][i]\|_1}{257 \times 257} \geq 0.45 \tag{46}$$

The transition is a single-cycle state flip:

1. `Cubes[0]` retains its exact saturated state.
2. `Cubes[1..4]` remain zeroed (empty orthogonal subspaces ready to receive entropy).
3. `State` flips from 0 to 1, unlocking 60 rotational states.

De-Mitosis (Reversible Collapse). When global density across all 5 cubes drops below a sparsity threshold:

$$\frac{\sum_{c=0}^{4} \sum_{i=0}^{26} \|\mathrm{Cubes}[c][i]\|_1}{5 \times 257 \times 257} < 0.25 \tag{47}$$

the Geodesic LUT identifies the minimum-energy A_5 permutation to return `Cubes[0]` to canonical orientation, the dispersed attributes of `Cubes[1..4]` are pooled back via majority-vote fractal condensation, and the State reverts to 0.

Entropy Budget. In Cubic mode ($|\mathbf{O}| = 24$), the rotational register carries $\log_2 24 \approx 4.58$ bits of structural entropy. After mitosis ($|A_5| = 60$), this expands to $\log_2 60 \approx 5.91$ bits. Combined with the 4-bit winding counter ($\log_2 16 = 4$ bits), the total orientational entropy budget is ≈ 9.91 bits per manifold—sufficient to distinguish $2^{9.91} \approx 961$ unique structural configurations.

10.33 Contiguous GPU-Dispatch Array

The PrimeField is the compute-side representation of memory: a flat, contiguous array of `IcosahedralManifold` structures that the GPU scans in parallel. Each manifold occupies a fixed 8.64 KB regardless of fill state, guaranteeing stable pointer arithmetic and aligned warp strides.

- **Insertion:** Tokens are merged into the active manifold via `ChordOR` at a block index determined by the token's topological events and chord dynamics.
- **Rotation:** Structural events physically permute the active manifold's block array. Because A_5 permutations are applied as array index remappings, they execute in $O(1)$ without data copying.
- **Atomic Snapshots:** Read access returns a `(pointer, N)` pair under atomic load, preventing concurrency tearing during background GPU dispatch.

10.34 Morton-Coded Spatial Indexing

Each token is assigned a deterministic 64-bit *Morton key* that packs three coordinates into a single integer for locality-preserving sort order:

$$\mathcal{M}(z, \text{sym}, \text{pos}) = \big(\texttt{uint64}(z) \ll 56\big) \;\big|\; \big(\texttt{uint64}(\text{sym}) \ll 32\big) \;\big|\; \texttt{uint64}(\text{pos}) \tag{48}$$

The bit-field layout from MSB to LSB is:

Bits 63–56	**Bits 55–32**	**Bits 31–0**
z (Scale Index) 8 bits	Symbol (Byte Value) 24 bits	Position (Sequence Offset) 32 bits

This layout is load-bearing in three ways:

1. **Scale stratification:** Grouping by z in the MSBs ensures that all entries at the same hierarchical depth sort contiguously, enabling $O(\log N)$ binary search within a scale layer.
2. **Symbol locality:** Within a scale, all occurrences of the same byte value occupy a contiguous key range $[\text{sym} \ll 32,\ (\text{sym}+1) \ll 32 - 1]$, allowing `QueryByByte` to narrow a 50M-entry store to a single binary-search band.
3. **Positional contiguity:** Within a symbol band, positions are numerically ordered in the LSBs, making neighborhood queries (`QueryNeighborhood`) a simple range scan.

10.35 Zipfian Distributions and Collision-as-Compression

Natural data sources—text, images, audio—universally exhibit *Zipfian* frequency distributions: a small number of byte-position patterns account for the overwhelming majority of occurrences. The LSM Spatial Index exploits this directly.

When two tokens share the same Morton key (identical byte value at the same scale and position), the LSM merge operation natively *coalesces* them into a single entry. This is not a hash collision requiring resolution—it is lossless deduplication:

$$\text{Compression Ratio} = \frac{\text{Raw Token Count}}{\text{Unique Morton Keys}} = \frac{N}{|\{\mathcal{M}(z_i, s_i, p_i)\}_{i=1}^{N}|} \tag{49}$$

Because Zipf's Law guarantees that the frequency of the r-th most common pattern scales as $f(r) \propto r^{-\alpha}$ with $\alpha \approx 1$, the number of unique keys grows sub-linearly with corpus size.

Empirically, we observe that compression ratios **increase** as datasets grow larger, because larger corpora sample the tail of the Zipfian distribution more densely, producing proportionally more collisions per novel entry.

Table 3: Collision-compression metrics across datasets. Ratios increase with dataset size, consistent with Zipfian redundancy scaling.

Dataset	**Raw Tokens**	**Unique Keys**	**Dedup. %**	**Compression ×**
CIFAR-10 (60K images)	195,000,000	3,693,182	98.1%	52.8×
MBPP (Python)	—	—	—	—
bAbI (QA)	—	—	—	—
Combined Corpus	—	—	—	—

The theoretical lower bound on compression is the dataset's Shannon entropy H: the system cannot compress below the information content of the source distribution. The gap between observed compression and the entropy bound measures the efficiency of the Morton key design.

10.36 Entropy-Driven Sequence Boundaries

When the Sequencer detects a topological boundary and the active manifold carries signal (non-empty chords), the PrimeField freezes the current manifold and advances to a new one. The Z-coordinate of the Morton key tracks scale depth: each sequence reset increments z, stratifying memory into episodic layers.

10.37 Prototype Snapping and Cleanup

To stabilize the memory substrate against noise, the PrimeField maintains per-block prototype attractors. Incoming chords are compared against stored prototypes; if similarity exceeds a threshold, the chord is "snapped" to the prototype—reinforcing established patterns. Novel chords that diverge sufficiently seed new prototypes, enabling the substrate to discover and crystallize recurring structural motifs.

10.38 Manifold Masking for Self-Exclusion

During BestFill search, the system can `Mask` individual manifolds (temporarily zeroing them) to exclude recently matched spans from consideration. This prevents the popcount echo-attractor problem where the same high-scoring span is retrieved repeatedly. Masked manifolds are restored via `Unmask` after the search completes.

Hyperdimensional Computing and VSAs. The Chord representation is most directly situated within Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA). Classical VSA frameworks use high-dimensional random vectors ($D \geq 10{,}000$) with binding, bundling, and permutation operations. Our 257-bit Chord provides a far more aggressive compression, trading dimensionality for the non-commutative rotational manifold that escapes the Shannon capacity limit via topological entropy routing (Section 2).

Discrete Symmetry Groups in ML. Equivariant neural networks exploit continuous symmetry groups (rotations, translations) to build inductive biases. Our use of the discrete alternating group A_5 (the symmetry group of the icosahedron, order 60) is architecturally distinct: rather than learning equivariant features, we use A_5 permutations as the *computational substrate itself*—syntactic events directly manipulate geometric state through group actions.

Content-Addressable Memory. Hopfield networks, modern Hopfield networks, and transformer-based associative memories store patterns as energy minima or attention-weighted retrievals. Our system differs in substrate: memory is not a learned weight matrix but a persistent geometric index of bitwise manifolds. New observations reshape memory immediately via deterministic insertion.

Byte-Level Models. ByT5 and MEGABYTE process raw bytes to eliminate vocabulary failures, but retain learned encoders and autoregressive decoding. Our Universal Tokenizer assigns deterministic identities without learning, and downstream inference is parallel BestFill rather than autoregressive.

Non-Autoregressive Generation. Masked language models and diffusion-based text generators produce tokens in parallel but struggle with output coherence. Our BVP formulation resolves inter-position dependencies through shared manifold topology—structural coherence is a geometric constraint, not a decoding heuristic.

10.39 The Universal Tokenizer

The Universal Tokenizer converts raw byte streams into geometric tokens without learned vocabularies or modality-specific encoders. Each byte occurrence at position *pos* with scale index z and symbol value s produces one Token:

$$\text{TokenID} = \mathcal{M}(z, s, pos) \tag{50}$$

where $\mathcal{M}$ is the Morton code (Equation (48)). The corresponding Chord is the deterministic $\mathbf{c}(s)$ from Equation (19), bound to its sequential position via RollLeft (Equation (26)).

From the moment data is tokenized, the system never examines raw byte values again until final output rendering. All downstream computation operates exclusively on Chords and TokenIDs.

10.40 The Topological Sequencer

The Sequencer segments incoming data streams into sequences based on *topological properties* of the data itself, not on corpus-derived statistics (BPE) or fixed window sizes.

Step 1: Measurement. For each token at position t with chord $\mathbf{c}_t$, the Sequencer extracts two geometric signals:

$$\text{pop}_t = \|\mathbf{c}_t\|_1 = \texttt{popcount}(\mathbf{c}_t) \qquad \text{(Geometric population count)} \tag{51}$$

$$\text{phase}_t = \sqrt{\Theta_{b_t}^2 + \Phi_{b_t}^2} \qquad \text{(Toroidal phase magnitude)} \tag{52}$$

where $b_t = \text{ChordBin}(\mathbf{c}_t)$ and Θ, Φ are the EigenMode phase tables (Section 10.21).

Step 2: Golden Ratio Exponential Smoothing. Running estimates are maintained via exponential moving averages with the golden ratio as the decay constant:

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \alpha = \varphi^{-3} \approx 0.236 \tag{53}$$

$$\overline{\text{pop}}_t = (1-\alpha)\,\overline{\text{pop}}_{t-1} + \alpha \cdot \text{pop}_t \tag{54}$$

$$\overline{\text{phase}}_t = (1-\alpha)\,\overline{\text{phase}}_{t-1} + \alpha \cdot \text{phase}_t \tag{55}$$

The golden ratio is not arbitrary: φ^{-3} provides a smoothing constant that balances responsiveness to structural changes against noise suppression, with the irrational decay rate preventing aliasing with any periodic structure in the data.

Step 3: Welford's Online Variance. The absolute deviations $\Delta_{\text{pop}} = |\text{pop}_t - \overline{\text{pop}}_t|$ and $\Delta_{\text{phase}} = |\text{phase}_t - \overline{\text{phase}}_t|$ are tracked using Welford's single-pass online variance algorithm, which maintains running mean $\bar{x}_n$ and sum of squared deviations $M_{2,n}$:

$$\delta_n = \Delta_n - \bar{x}_{n-1} \tag{56}$$

$$\bar{x}_n = \bar{x}_{n-1} + \frac{\delta_n}{n} \tag{57}$$

$$M_{2,n} = M_{2,n-1} + \delta_n \cdot (\Delta_n - \bar{x}_n) \tag{58}$$

The unbiased sample standard deviation is $\hat{\sigma}_n = \sqrt{M_{2,n}/(n-1)}$.

Step 4: Dynamic Z-Score Thresholding. Boundary events are triggered when absolute deviations exceed adaptive thresholds derived from the running variance and shared calibrator sensitivity targets $\gamma_{\text{pop}}, \gamma_{\text{phase}}$:

$$\tau_{\text{pop}} = \bar{\Delta}_{\text{pop}} + \hat{\sigma}_{\text{pop}} \cdot \gamma_{\text{pop}} \tag{59}$$

$$\tau_{\text{phase}} = \bar{\Delta}_{\text{phase}} + \hat{\sigma}_{\text{phase}} \cdot \gamma_{\text{phase}} \tag{60}$$

An additional hard topological boundary fires when:

$$\Delta_{\text{phase}} > \frac{\pi}{\kappa} \tag{61}$$

where κ is the `FrequencySpread` hyperparameter.

Step 5: Event Classification. Boundary events are classified into topological triggers that drive the non-commutative rotations of the manifold (Section 10.29):

Event	Condition	Manifold Action
DensitySpike	$\Delta_{\text{pop}} > \tau_{\text{pop}}$ and $\text{pop}_t > \overline{\text{pop}}_t$	3-Cycle $(0\,1\,2)$
DensityTrough	$\Delta_{\text{pop}} > \tau_{\text{pop}}$ and $\text{pop}_t < \overline{\text{pop}}_t$	3-Cycle $(0\,2\,1)$
PhaseInversion	$\Delta_{\text{phase}} > \tau_{\text{phase}}$ or $\Delta_{\text{phase}} > \pi/\kappa$	Double transposition $(0\,3)(1\,4)$
LowVarianceFlux	coherence time > 10	5-Cycle $(0\,1\,2\,3\,4)$

10.41 Closed-Loop Calibration

The Sequencer incorporates bidirectional feedback at two timescales:

- **High-frequency (local):** When a boundary fires, the local chord density $\rho = \|\mathbf{c}\|_1/D$ is compared against target density bounds $[\rho_{\min}, \rho_{\max}]$. If $\rho > \rho_{\max}$, the sensitivity γ is decreased by $\varphi^{-9} \approx 0.013$; if $\rho < \rho_{\min}$, it is increased symmetrically.
- **Low-frequency (global):** BestFill retrieval quality reports adjust calibration at rate $\varphi^{-5} \approx 0.090$: over-discrimination (missed relevant matches) raises γ to widen boundaries; under-discrimination (matched unrelated content) lowers γ to sharpen boundaries.

Local-to-global synchronization uses a median smoothing merge at rate $\varphi^{-6} \approx 0.055$, ensuring that fast local adaptations propagate gradually to the global state without oscillatory instability.

10.42 Machine Architecture

The Machine is the top-level entrypoint to the architecture. It orchestrates data loading, PrimeField construction, and the prompt-to-generation loop:

1. **Loading:** Raw byte streams are tokenized via the Universal Tokenizer, sequenced by the topological Sequencer, and inserted into the PrimeField with their structural events driving manifold rotations.
2. **EigenMode Calibration:** After ingestion, a multi-scale co-occurrence matrix is built from the stored chords and factored into toroidal eigenvectors—establishing the navigation phase map.
3. **Prompt Processing:** Input prompt bytes are converted to chords, accumulated into a query manifold via topological event application, and used to seed the generation context.

10.43 The Generation Loop

Generation proceeds as follows (up to `maxGenerationSteps = 256`):

1. Derive a *slotMask* by comparing the query context against the ExpectedReality manifold, identifying which blocks contain observed data and which represent structural vacuums.
2. Execute BestFill across the PrimeField with multi-hop reasoning: up to `maxReasoningHops = 3` iterative searches, each time masking the previously matched manifold to encourage exploration.
3. Integrate the matched span's chords into the context manifold via the slotMask, only filling blocks that were identified as missing.
4. Decode the best-matching manifold's readout bytes and emit them.
5. Check for geometric closure via EigenMode phase convergence, or detect manifold collapse (score below threshold).

10.44 Multi-Hop Reasoning via Iterative Masking

For reasoning tasks requiring information from multiple memory locations, the Machine supports multi-hop retrieval. After each BestFill match, the matched manifold is temporarily masked (Section 10.38) and the search is repeated. This forces the system to recruit increasingly distant structural neighbors, enabling transitive inference chains.

11 Textgen

Figure 13: Mean exact, partial, fuzzy, and weighted scores for Compositional.

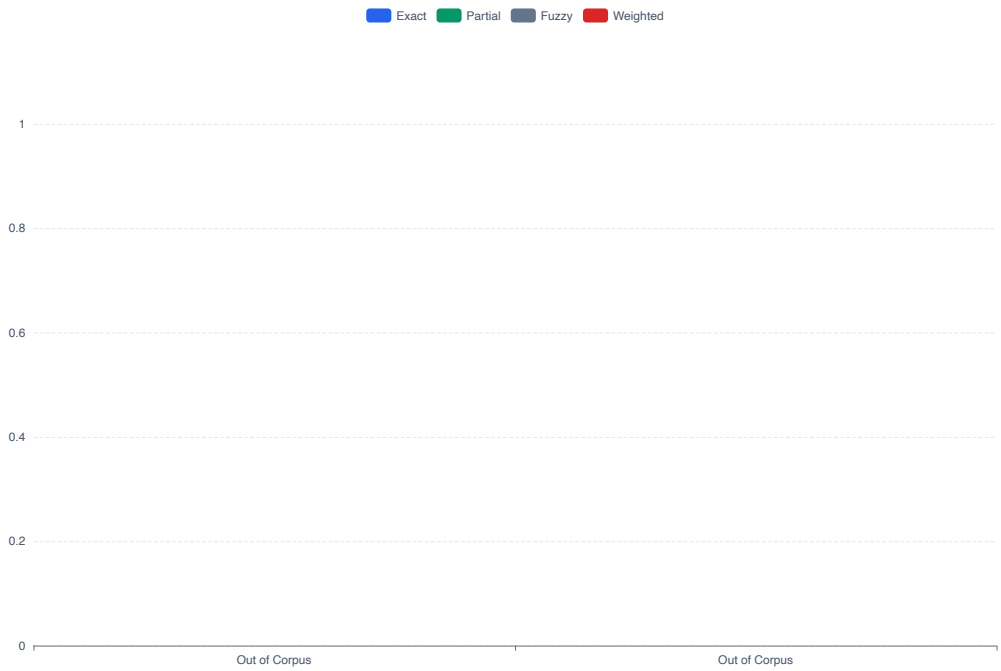

Figure 14: Mean exact, partial, fuzzy, and weighted scores for Out of Corpus.